

Project Data Page:

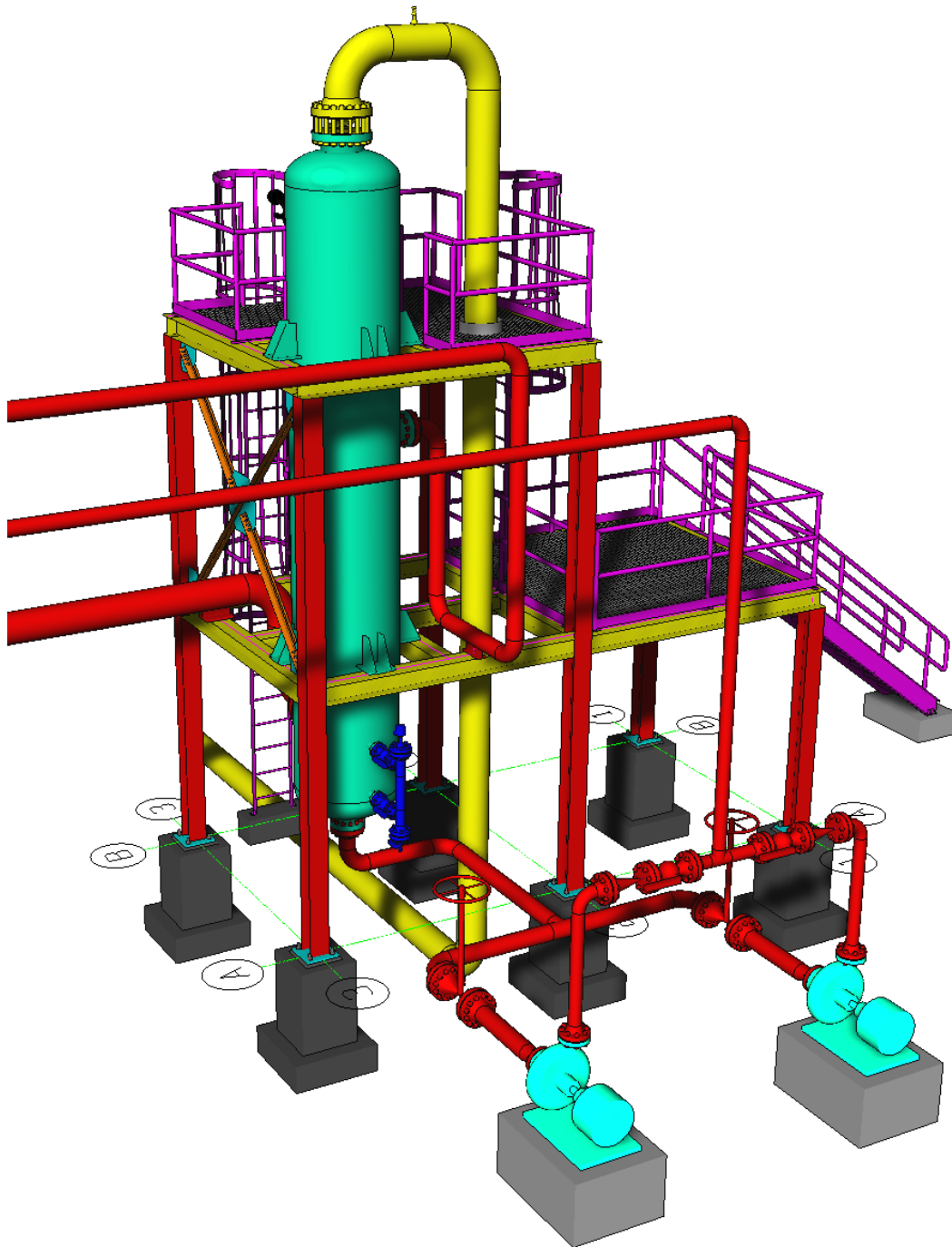




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\401-R-31001Rev 00.pvdb

Date of Analysis : May 25, 2026 5:34pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 2500.000 mm OD
 Vessel Design Length, Tangent to Tangent 2454.32 cm
 Distance of Bottom Tangent above Grade 900.00 cm
 Distance of Base above Grade 900.00 cm
 Specified Datum Line Distance 0.00 cm
 Internal Design Temperature 300 °C
 Internal Design Pressure 163.155 kgf/cm²
 External Design Temperature 300 °C
 External Design Pressure 1.055 kgf/cm²
 Maximum Allowable Working Pressure 164 kgf/cm²
 External Max. Allowable Working Pressure 82 kgf/cm²
 Shop Test Pressure 260.871 kgf/cm²
 Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature 11.0 °C
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code Response Spectrum

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	+IT
Flange	SA-336 F11	2	...	K11572	+IT
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-336 F22	3	...	K21590	
Nozzle Flg	SA-182 F22	3	...	K21590	
Nozzle Flg	SA-182 F11	2	...	K11572	
Basering	SA-516 70	...	...	K02700	
Flg Bolting	SA-540 B23	3	<= 240	H43400	
Base Bolting	SA-193 B7	...	<= 64	G41400	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom head	163.234	1.05	166.000	3.0000	N/A	No
Shell-1	163.230	1.05	164.000	3.0000	N/A	No
Shell-2	163.221	1.05	164.000	3.0000	N/A	No
Inte skirt	...	...	...	...	N/A	No
Shell-3	163.216	1.05	164.000	3.0000	N/A	No
Shell-4	163.206	1.05	164.000	3.0000	N/A	No
Shell-5	163.197	1.05	164.000	3.0000	N/A	No
Shell-6	163.188	1.05	164.000	3.0000	N/A	No
Shell-7	163.179	1.05	164.000	3.0000	N/A	No
Shell-8	163.169	1.05	164.000	3.0000	N/A	No
Shell flange-1	163.162	1.05	164.000	3.0000	N/A	No
Shell flange-2	163.157	1.05	175.000	3.0000	N/A	No

Liquid Level: 2579.80 cm Dens.: 30.720 kg/m³ Sp. Gr.: 0.031



Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Sphere	0.0	0.0	72.0	67.0	66.1	6.6	1.00	1.00
Cylinder	300.0	300.0	132.0	132.0	131.5	23.6	1.00	1.00
Cylinder	485.0	185.0	132.0	132.0	131.5	23.6	1.00	1.00
Skirt	485.0	300.0	30.0	30.0	...	...	1.00	1.00
Cylinder	785.0	300.0	132.0	132.0	131.5	23.6	1.00	1.00
Cylinder	1085.0	300.0	132.0	132.0	131.5	23.6	1.00	1.00
Cylinder	1385.0	300.0	132.0	132.0	131.4	23.6	1.00	1.00
Cylinder	1685.0	300.0	132.0	132.0	131.4	23.6	1.00	1.00
Cylinder	1985.0	300.0	132.0	132.0	131.4	23.6	1.00	1.00
Cylinder	2240.0	255.0	132.0	132.0	131.4	23.6	1.00	1.00
Body Flg	2400.3	160.0	1380.0	1375.0	219.4	219.4	1.00	1.00
Body Flg	2454.3	54.0	630.0	540.0	520.8	217.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
B 450mm	WRC-107/537	0.990	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	268.174	...	647.681	...	0.33820E-08
20	30	35	656.643	3766.93	546.156	1.0282
30	40	207.5	2524.83	8050.12	5318.37	11301.8
40	50	150	19907.4	46736.1	186599	828954
50	60	450	14060.4	46443.7	141155	689905
60	70	750	12046.8	45432.4	101994	551930
70	80	1050	9987.23	43789.6	68942.8	417013
80	90	1350	7870.96	38510.9	42155.5	287281
90	100	1650	5695.66	31539.7	21805.5	172816
100	110	1927.5	3460.52	22737.9	8071.27	78616.3
110	120	2135	1524.92	11491.7	1709.99	20685.5
120	130	2242	457.995	4258.09	123.659	2299.38

Abs Max of the all of the Stress Ratio's : 0.9376

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	38.0000	mm
Inside Diameter of Basing	3603.0002	mm
Outside Diameter of Basing	4008.9998	mm
Nominal Diameter of Bolts	42.0000	mm
Diameter of Bolt Circle	3895.0002	mm
Number of Bolts	16	

Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	102.0000	mm
Height of Gussets	204.0000	mm
Distance between Gussets	102.0000	mm
Thickness of Top Plate or Ring	38.0000	mm
Radial Width of the Top Plate	102.0000	mm

Loads for Foundation/Support Design:

Total Wind Shear on Support	19907.	kgf
Total Earthquake Shear on Support	46736.	kgf
Wind Moment on Support	193857.	kgf-m.
Earthquake Moment on Support	836212.	kgf-m.



Maximum Initial Bolt Load 18233. kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	309146.0	kg
Shop Test - Fabricated + water (full)	431127.7	kg
Shipping - Fab. + rem. intls.+ shipping app.	309146.0	kg
Erected - Fab. + rem. intls.+ insul. (etc)	563282.7	kg
Empty - Fab. + intls. + details + wghts.	563282.7	kg
Operating - Empty + operating liquid (no ca)	567033.1	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
B 450mm	166	OK	...	OK	OK	Passed
Top plug weight	561	OK	...	OK	Failed	Passed
AT 2"	1444	OK	...	OK	OK	Passed
BT 2"	1444	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 166 Nozzle : B 450mm
 Minimum MAWP Shells/Flanges : 164 Element : Shell-1
 Minimum MAPnc Shells/Flanges : 176 Element : Shell flange-1

Computed Vessel M.A.W.P. : 164 kgf/cm²

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
10	B 450mm	0.000	0.000	228.000
120	Top plug weight	0.000	0.000	28.000
120	AT 2"	0.000	30.000	28.400
120	BT 2"	0.000	210.000	28.400

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
Top plug weight	50 mm	Actual	None	72.0	11.000	...	...	565	...
AT 2"	2 in	LWN	RTJWn	66.8	8.000	...	...	1540	1500
BT 2"	2 in	LWN	RTJWn	66.8	8.000	...	...	1540	1500
B 450mm	450 mm	Actual	WNF	522.0	36.000	...	...	918	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Top plug we	SA-336 F22	11.000	9.000	...	...	...
AT 2"	SA-336 F22	530.000	10.000	...	...	...
BT 2"	SA-336 F22	530.000	10.000	...	...	...
B 450mm	SA-336 F22	67.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Top plug weight	...	0.0	25.00	0.00	Shell flange-2
AT 2"	...	30.0	1000.00	0.00	Shell flange-2
BT 2"	...	210.0	1000.00	0.00	Shell flange-2
B 450mm	...	0.0	800.00	0.00	Bottom head

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 72.0mm THK X 2516.0mm ID X 0.0cm	SA-387 11
1	WELD NECK FLANGE: 1380.0mm THK X 3064.0mm OD	SA-336 F11
1	BLIND FLANGE: 630.0mm THK X 3070.0mm OD	SA-336 F11
1	INSULATION: 1258mm X 120mm THK	
6	INSULATION: 3000mm X 120mm THK	
1	INSULATION: 1850mm X 120mm THK	
1	INSULATION: 2550mm X 120mm THK	
1	INSULATION: 765mm X 120mm THK	
1	INSULATION: 540mm X 120mm THK	
1	BASERING: 4009.0mm OD X 3603.0 ID X 38.0mm	SA-516 70
16	BASERING BOLTS: 42.0mm DIA	SA-193 B7
16	BASERING BOLT NUTS: 42.0mm DIA	...
1	BASERING TOP RING: 2999.2mm OD X 2795.2 ID X 38.0mm	...
32	BASERING GUSSET PLATES: 102.0mm X 102.0mm X 204.0mm X 16.0mm	...
1	GASKET: 2555.0mm OD X 2530.0mm ID	...
28	BODY FLANGE BOLTS: 150.0mm DIA	SA-540 B23
56	NUTS FOR BODY FLANGE BOLTS: 150.0mm DIA	...
2	CYLINDRICAL SEGMENT 132.0mm THK X 3119.7mm X 300.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 132.0mm THK X 1614.4mm X 300.0cm	SA-387 11
2	CYLINDRICAL SEGMENT 132.0mm THK X 3048.0mm X 185.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 132.0mm THK X 1757.9mm X 185.0cm	SA-387 11
1	CON SKIRT SEGMENT 30.0mm THK X 3341.6mm X 300.0cm X 4537.7mm	SA-516 70
1	CON SKIRT SEGMENT 30.0mm THK X 3317.8mm X 300.0cm X 4505.3mm	SA-516 70
1	CON SKIRT SEGMENT 30.0mm THK X 1933.4mm X 300.0cm X 2625.4mm	SA-516 70
5	CYLINDRICAL SEGMENT 132.0mm THK X 3048.0mm X 300.0cm	SA-387 11
5	CYLINDRICAL SEGMENT 132.0mm THK X 4805.9mm X 300.0cm	SA-387 11
5	CYLINDRICAL SEGMENT 132.0mm THK X 0.0mm X 300.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 132.0mm THK X 4805.9mm X 255.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 132.0mm THK X 3048.0mm X 255.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 132.0mm THK X 0.0mm X 255.0cm	SA-387 11
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Shell flange-[11]	!	11		Yes			
Shell flange-[11]	!	11		Yes			
Bottom head [9]	!	11		Yes			
Shell-1 [8]	!	11		Yes			
Shell-2 [8]	!	11		Yes			
Shell-3 [8]	!	11		Yes			
Shell-4 [8]	!	11		Yes			
Shell-5 [8]	!	11		Yes			
Shell-6 [8]	!	11		Yes			
Shell-7 [8]	!	11		Yes			
Shell-8 [8]	!	11		Yes			
B 450mm [1]	B	14	2	Yes	0.780	67.000	1.00
Nozzle Flg [5]	B	-18	-39	Yes			
Top plug weigh[1]	!	11		Yes			
AT 2" [1]	!	11		Yes			
Nozzle Flg [5]	B	-18	-39	Yes			
BT 2" [1]	!	11		Yes			
Nozzle Flg [5]	B	-18	-39	Yes			
Warmest MDMT:		14	2				
Required Minimum Design Metal Temperature				11.4 °C			
Warmest Computed Minimum Design Metal Temperature				11.0 °C			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (Kgfc/m2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	163.15	kgf/cm ²
Design Internal Temperature	300.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	11.4	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	User Defined Hydro
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	900	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class B
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1040	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/12
Element From Node	10



Element To Node	20
Element Type	Spherical
Description	Bottom head
Distance "FROM" to "TO"	0 cm
Inside Diameter	2516 mm
Element Thickness	67 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	72 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.15 kgf/cm ²
Design Temperature Internal Pressure	300 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.33
Material Name	SA-387 11 +IT
Allowable Stress, Ambient	2110.7 kgf/cm ²
Allowable Stress, Operating	1672.3 kgf/cm ²
Allowable Stress, Hydrotest	3002.9 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2508.4 kgf/cm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2 ::
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-125.8 cm
Height/Length of Liquid	125.8 cm
Liquid Density	30.72 kg/m ³

Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 90
Dist. from "FROM" Node / Offset dist	-125.8 cm
Height/Length of Insulation	125.8 cm
Thickness of Insulation	120 mm
Density	0.00015 kg/cm ³

Element From Node	10
Detail Type	Nozzle
Detail ID	B 450mm
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	450 mm
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1538.1 kgf
Grade of Attached Flange	GR 1.10
Nozzle Matl	SA-336 F22

Element From Node	10
Detail Type	Weight
Detail ID	SF weight 1
Dist. from "FROM" Node / Offset dist	0 cm
Miscellaneous Weight	800 kgf
Offset from Element Centerline	1300 mm

Element From Node	10
Detail Type	Weight
Detail ID	SF weight 2
Dist. from "FROM" Node / Offset dist	0 cm
Miscellaneous Weight	800 kgf



Offset from Element Centerline 1300 mm

Element# 2/12
 Element From Node 20
 Element To Node 30
 Element Type Cylinder
 Description Shell-1
 Distance "FROM" to "TO" 300 cm
 Inside Diameter 2500 mm
 Element Thickness 132 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 132 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 163.15 kgf/cm²
 Design Temperature Internal Pressure 300 °C
 Design External Pressure 1.0546 kgf/cm²
 Design Temperature External Pressure 300 °C
 Effective Diameter Multiplier 1.33
 Material Name SA-387 11 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid: 20
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 300 cm
 Liquid Density 30.72 kg/m³

Element From Node 20
 Detail Type Insulation
 Detail ID Ins: 90
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 300 cm
 Thickness of Insulation 120 mm
 Density 0.00015 kg/cm³

Element From Node 20
 Detail Type Weight
 Detail ID Support ring1
 Dist. from "FROM" Node / Offset dist 0 cm
 Miscellaneous Weight 2999.9 kgf
 Offset from Element Centerline 1460 mm

Element From Node 20
 Detail Type Weight
 Detail ID Support ring2
 Dist. from "FROM" Node / Offset dist 0 cm
 Miscellaneous Weight 2999.9 kgf
 Offset from Element Centerline 1460 mm

Element From Node 20
 Detail Type Weight
 Detail ID Tailing lug
 Dist. from "FROM" Node / Offset dist 220 cm
 Miscellaneous Weight 3499.9 kgf
 Offset from Element Centerline 1850 mm

Element# 3/12
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description Shell-2
 Distance "FROM" to "TO" 185 cm
 Inside Diameter 2500 mm



Element Thickness	132	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	132	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.15	kgf/cm ²
Design Temperature Internal Pressure	300	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	185	cm
Liquid Density	30.686	kg/m ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	185	cm
Thickness of Insulation	120	mm
Density	0.00015	kg/cm ³

Element From Node	30	
Detail Type	Weight	
Detail ID	Catalyst-3	
Dist. from "FROM" Node / Offset dist	185	cm
Miscellaneous Weight	97750	kgf
Offset from Element Centerline	0	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Skirt band-1	
Dist. from "FROM" Node / Offset dist	185	cm
Miscellaneous Weight	3000	kgf
Offset from Element Centerline	1400	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Skirt band-2	
Dist. from "FROM" Node / Offset dist	185	cm
Miscellaneous Weight	3000	kgf
Offset from Element Centerline	1400	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Basket-3	
Dist. from "FROM" Node / Offset dist	185	cm
Miscellaneous Weight	25365	kgf
Offset from Element Centerline	0	mm

Element#	4/12	
Element From Node	40	
Element To Node	50	
Element Type	Skirt Support	
Description	Inte skirt	
Distance "FROM" to "TO"	300	cm
Skirt Outside Diameter	2796	mm
Diameter of Skirt at Base	3775	mm
Skirt Thickness	30	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	30	mm



External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	300	°C
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-516 70	
Allowable Stress, Ambient	1784.4	kgf/cm²
Allowable Stress, Operating	1386.8	kgf/cm²
Allowable Stress, Hydrotest	2518.6	kgf/cm²
Material Density	0.00775	kg/cm³
P Number Thickness	0	mm
Yield Stress, Operating	2080.2	kgf/cm²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Head-to-Skirt or Circ. Seam	1.0	
Weld is pre-Heated	No	

Element#	5/12	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	300	cm
Inside Diameter	2500	mm
Element Thickness	132	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	132	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.15	kgf/cm²
Design Temperature Internal Pressure	300	°C
Design External Pressure	1.0546	kgf/cm²
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-387 11 +IT	
Allowable Stress, Ambient	2110.7	kgf/cm²
Allowable Stress, Operating	1672.3	kgf/cm²
Allowable Stress, Hydrotest	3002.9	kgf/cm²
Material Density	0.00775	kg/cm³
P Number Thickness	0	mm
Yield Stress, Operating	2508.4	kgf/cm²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	300	cm
Liquid Density	30.686	kg/m³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	120	mm
Density	0.00015	kg/cm³



Element#	6/12
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	300 cm
Inside Diameter	2500 mm
Element Thickness	132 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	132 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.15 kgf/cm ²
Design Temperature Internal Pressure	300 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.33
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	300 cm
Liquid Density	30.72 kg/m ³

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 90
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	300 cm
Thickness of Insulation	120 mm
Density	0.00015 kg/cm ³

Element#	7/12
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell-5
Distance "FROM" to "TO"	300 cm
Inside Diameter	2500 mm
Element Thickness	132 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	132 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.15 kgf/cm ²
Design Temperature Internal Pressure	300 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.33
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	300 cm
Liquid Density	30.8 kg/m ³

Element From Node	70
Detail Type	Insulation
Detail ID	Ins: 90
Dist. from "FROM" Node / Offset dist	0 cm



Height/Length of Insulation	300	cm
Thickness of Insulation	120	mm
Density	0.00015	kg/cm ³

Element#	8/12	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	300	cm
Inside Diameter	2500	mm
Element Thickness	132	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	132	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.15	kgf/cm ²
Design Temperature Internal Pressure	300	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	300	cm
Liquid Density	30.786	kg/m ³

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	120	mm
Density	0.00015	kg/cm ³

Element From Node	80	
Detail Type	Weight	
Detail ID	Catalyst-2	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	41950	kgf
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Basket-2	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	25365	kgf
Offset from Element Centerline	0	mm

Element#	9/12	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	Shell-7	
Distance "FROM" to "TO"	300	cm
Inside Diameter	2500	mm
Element Thickness	132	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	132	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.15	kgf/cm ²



Design Temperature Internal Pressure	300	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	300	cm
Liquid Density	30.786	kg/m ³

Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	120	mm
Density	0.00015	kg/cm ³

Element#	10/12	
Element From Node	100	
Element To Node	110	
Element Type	Cylinder	
Description	Shell-8	
Distance "FROM" to "TO"	255	cm
Inside Diameter	2500	mm
Element Thickness	132	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	132	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.15	kgf/cm ²
Design Temperature Internal Pressure	300	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	100	
Detail Type	Liquid	
Detail ID	Liquid: 100	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	255	cm
Liquid Density	30.786	kg/m ³

Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	255	cm
Thickness of Insulation	120	mm
Density	0.00015	kg/cm ³

Element From Node	100	
Detail Type	Weight	
Detail ID	Cleat weight	
Dist. from "FROM" Node / Offset dist	99.997	cm
Miscellaneous Weight	2252	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
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Detail Type	Weight	
Detail ID	Catalyst-1	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	25750	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Basket-1	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	25365	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Trunnion-1	
Dist. from "FROM" Node / Offset dist	145	cm
Miscellaneous Weight	3500	kgf
Offset from Element Centerline	1382	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Trunnion-2	
Dist. from "FROM" Node / Offset dist	145	cm
Miscellaneous Weight	3500	kgf
Offset from Element Centerline	1382	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Elec heater	
Dist. from "FROM" Node / Offset dist	255	cm
Miscellaneous Weight	8455	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Nozzle-A	
Dist. from "FROM" Node / Offset dist	255	cm
Miscellaneous Weight	850	kgf
Offset from Element Centerline	1350	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Nozzle-C	
Dist. from "FROM" Node / Offset dist	255	cm
Miscellaneous Weight	270	kgf
Offset from Element Centerline	1350	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Nozzle-D	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	270	kgf
Offset from Element Centerline	1350	mm

Element#	11/12	
Element From Node	110	
Element To Node	120	
Element Type	Flange	
Description	Shell flange-1	
Distance "FROM" to "TO"	160	cm
Flange Inside Diameter	2500	mm
Element Thickness	1375	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	1380	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.15	kgf/cm ²
Design Temperature Internal Pressure	300	°C



Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-336 F11	+IT
Allowable Stress, Ambient	1876.2	kgf/cm ²
Allowable Stress, Operating	1488.7	kgf/cm ²
Allowable Stress, Hydrotest	2663.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2233.1	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K11572	
Class / Thickness / Grade	2 ::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Liquid	
Detail ID	Liquid: 110	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	160	cm
Liquid Density	30.72	kg/m ³

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	76.5	cm
Thickness of Insulation	120	mm
Density	0.00015	kg/cm ³

Element#	12/12	
Element From Node	120	
Element To Node	130	
Element Type	Flange	
Description	Shell flange-2	
Distance "FROM" to "TO"	54	cm
Flange Outside Diameter	3070	mm
Element Thickness	540	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	630	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.15	kgf/cm ²
Design Temperature Internal Pressure	300	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-336 F11	+IT
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	120	
Detail Type	Liquid	
Detail ID	Liquid: 120	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	54	cm
Liquid Density	30.72	kg/m ³

Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	54	cm
Thickness of Insulation	120	mm
Density	0.00015	kg/cm ³



Element From Node	120	
Detail Type	Nozzle	
Detail ID	Top plug weight	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1620	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22 +IT	

Element From Node	120	
Detail Type	Nozzle	
Detail ID	AT 2"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	30.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	63.341	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F22 +IT	

Element From Node	120	
Detail Type	Nozzle	
Detail ID	BT 2"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	210.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	63.341	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F22 +IT	



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Bottom head		...	...	...	...
Shell-1		...	300	...	300
Shell-2		...	485	...	185
Inte skirt		...	485	...	-300
Shell-3		...	785	...	300
Shell-4		...	1085	...	300
Shell-5		...	1385	...	300
Shell-6		...	1685	...	300
Shell-7		...	1985	...	300
Shell-8		...	2240	...	255
Shell flange-1		...	2400.32	...	160
Shell flange-2		...	2454.32	...	54

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Warnings for Flange : Shell flange :

Note: Flange Hub Taper 0.3622 is > 1:3. Check Fig. 4.16.2 of VIII-2 Part 4.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-2 in Table 1.1 or for non-Code vessels.

Flange Input Data Values

Description: Shell flange :

Shell flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		300	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2500.000	mm
Flange Outside Diameter	A	3064.000	mm
Flange Thickness	t	1375.0000	mm
Thickness of Hub at Small End	go	149.0000	mm
Thickness of Hub at Large End	gl	230.5000	mm
Length of Hub	h	225.0000	mm
Flange Material		SA-336 F11 +IT	
Flange Material UNS number		K11572	
Flange Allowable Stress At Temperature	Sfo	1488.73	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1876.18	kgf/cm ²
Bolt Material		SA-540 B23	
Bolt Allowable Stress At Temperature	Sb	2039.32	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	2039.32	kgf/cm ²
Diameter of Bolt Circle	C	2811.000	mm
Nominal Bolt Diameter	a	150.0000	mm
Type of Threads	Special,	Root Area	
Root Area of a Single Bolt		158.2800	cm ²
Number of Bolts		28	
Flange Face Outside Diameter	Fod	2555.000	mm
Flange Face Inside Diameter	Fid	2500.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2555.000	mm
Gasket Inside Diameter	Gi	2530.000	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (163.162(2500.0 + 3.0) / (1488.73 - 0.5 \cdot 163.162)) + (3.0)$$

$$= 148.2877 \text{ mm}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 1488.73 \cdot \log((2 \cdot 146.0 + 1253.0) / 1253.0)$$

$$= 164.083 \text{ kgf/cm}^2$$

Small End Hub MAPnc:

$$= Sa \cdot \log((2.0 \cdot T + R) / R)$$

$$= 1876.18 \cdot \log((2.0 \cdot 149.0 + 1250.0) / 1250.0)$$

$$= 211.284 \text{ kgf/cm}^2$$

The hub small end thickness is now taken as the attached shell thickness.

Corroded Flange Thickness, tc = T-ci	1372.000 mm
Corroded Flange ID, Bcor = B+2*ci	2506.000 mm
Corroded Large Hub, g1Cor = g1-ci	227.500 mm
Corroded Small Hub, g0Cor = g0-ci	129.000 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	-75.000 mm
Gasket Contact Width, N = Self-Energizing	0.000 mm
Basic Gasket Width, bo = N/2	0.000 mm
Effective Gasket Width, b = bo	0.000 mm
Gasket Reaction Diameter, G = Go (Self-Energizing)	2555.000 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2555.0^2 \cdot 163.162$$

$$= 8365444.000 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 0.0 \cdot \pi \cdot 2555.0 \cdot 0.0 \cdot 163.16$$

$$= 0.000 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 2506.0^2 \cdot 163.1616$$

$$= 8047654.500 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 8365444 - 8047655$$

$$= 317789.281 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(8365444 + 0 + 0, 0)$$

$$= 8365444.000 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 0.0 \cdot 0.0 \cdot \pi \cdot 2555.0 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 0.000 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((8365444 + 0 + 4 \cdot 0/2555.0)/2039, 0/2039)$$

$$= 4102.071 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 150.0 + 6 \cdot 1372.0/(0.0 + 0.5)$$

$$= 16764.000 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2811.0 \cdot \sin(\pi/28)$$

$$= 314.732 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs/(2a + t) }, 1)$$

$$= \max(\sqrt{ 314.732/(2 \cdot 150.0 + 1372.0) }, 1)$$

$$= 1.0000$$

Bolting Information for User Defined Bolts (Non Mandatory):

	Minimum	Actual	Maximum



Bolt Area: 4102.071 4431.840
 Circ. Spacing between the Bolts: 0.000 314.732 16764.000

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 2039.32 \cdot 4431.8403$$

$$= 9037949.00 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 8365444 - 8365444$$

$$= 0.00 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2811.0 - 2555.0) / 2$$

$$= 128.0000 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((2811.0 - 2506.0) / 2 + 128.0)$$

$$= 140.2500 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (2811.0 - 2506.0 - 227.5) / 2$$

$$= 38.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8047654.	38.7500	1.0000	311847.
Face Pressure, Mt	317789.	140.2500	1.0000	44570.
Gasket Load, Mg	0.	128.0000	1.0000	0.
Gasket Seating, Matm	9037949.	128.0000	1.0000	1156859.

Total Moment for Operation, Mo 356417. kgf-m.
 Total Moment for Gasket seating, Mg 1156859. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 568.572 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.396
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 1.764

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.853
 Factor V per 4.16.5 0.292
 Factor f per 4.16.5 1.257

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 3064.0 / 2506.0$$

$$= 1.223$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.223^2 \cdot \log_{10}(1.223))) / (1.223^2 - 1)) / (1.223 - 1)$$

$$= 9.773$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.223^2 (1 + 8.55246 \cdot \log_{10}(1.223)) - 1) / ((1.0472 + 1.9448 \cdot 1.223^2) \cdot (1.223 - 1))$$

$$= 1.830$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.223^2 (1 + 8.55246 \cdot \log_{10}(1.223)) - 1) / (1.36136 (1.223^2 - 1) \cdot (1.223 - 1))$$

$$= 10.739$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.223^2 + 1) / (1.223^2 - 1)$$

$$= 5.041$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 10.739 \cdot 129.0^2 \cdot 568.572 / 0.292$$

$$= 348070976.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.853 / 568.572$$

$$= 0.002 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (1372.0 \cdot 0.002 + 1) / 1.83 + 1372.0^3 / 348070976.0$$

$$= 9.091$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.2567 \cdot 356417 / 2506.0) / (9.0913 \cdot 227.5^2)$$

$$= 37.99 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.2567 \cdot 1156859 / 2506.0) / (9.0913 \cdot 227.5^2)$$

$$= 123.30 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (3.7445 \cdot 356417 / 2506.0) / (9.0913 \cdot 1372.0^2)$$

$$= 3.11 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (3.7445 \cdot 1156859 / 2506.0) / (9.0913 \cdot 1372.0^2)$$

$$= 10.10 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (9.773 \cdot 356417 / (1372.0^2 \cdot 2506.0)) - 5.0411 \cdot 3$$

$$= 58.15 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (Y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (9.773 \cdot 1156859 / (1372.0^2 \cdot 2506.0)) - 5.0411 \cdot 10$$

$$= 188.75 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (38 + \max(3, 58)) / 2$$

$$= 48.07 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (123 + \max(10, 189)) / 2$$

$$= 156.02 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 8365444 / 4431.8403$$

$$= 1887.58 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
 = (0/4431.8403)
 = 0.00 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	37.99	2233.10	123.30	2814.27
Radial Flange	3.11	1488.73	10.10	1876.18
Tangential Flange	58.15	1488.73	188.75	1876.18
Maximum Average	48.07	1488.73	156.02	1876.18
Bolting	1887.58	2039.32	0.00	2039.32

Minimum Required Flange Thickness 219.380 mm

Estimated M.A.W.P. (Operating) 164 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 28927.6 kg
 Estimated Unfinished Weight of Forging at given Thk 30563.2 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 1156859 / 1.0 \cdot 100000.023 \cdot 0.292 / (0.753 \cdot 2081087 \cdot 129.0^2 \cdot 568.572 \cdot 0.3)$
 = 0.395 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 356416.9 / 1.0 \cdot 100000.023 \cdot 0.292 / (0.753 \cdot 1901117 \cdot 129.0^2 \cdot 568.572 \cdot 0.3)$
 = 0.133 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 1156859 / 1.0 \cdot 100000.023 \cdot 0.292 / (9.091 \cdot 2081087 \cdot 129.0^2 \cdot 568.572 \cdot 0.3)$
 = 0.032 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 356416.9 / 1.0 \cdot 100000.023 \cdot 0.292 / (9.091 \cdot 1901117 \cdot 129.0^2 \cdot 568.572 \cdot 0.3)$
 = 0.011 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	39.26	2814.27	121.79	2814.27
Radial Flange	3.30	1876.18	10.24	1876.18
Tangential Flange	59.76	1876.18	185.40	1876.18
Maximum Average	49.51	1876.18	153.60	1876.18
Bolting	1887.58	2039.32	0.00	2039.32

Minimum Req'd Thickness, ambient, no Corrosion: 216.637 mm

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Flange Input Data Values

Description: New Flange :

Shell flange-2

Flange Type		Blind	
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		300	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	3070.000	mm
Flange Thickness	t	540.0000	mm
Flange Material		SA-336 F11 +IT	
Flange Material UNS number		K11572	
Flange Allowable Stress At Temperature	Sfo	1488.73	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1876.18	kgf/cm ²
Bolt Material		SA-540 B23	
Bolt Allowable Stress At Temperature	Sb	2039.32	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	2039.32	kgf/cm ²
Perimeter along the Center of the Bolts	L	8831.018	mm
Diameter of Bolt Circle	C	2811.000	mm
Nominal Bolt Diameter	a	150.0000	mm
Type of Threads	Special,	Root Area	
Root Area of a Single Bolt		158.2800	cm ²
Number of Bolts		28	
Flange Face Outside Diameter	Fod	2555.000	mm
Flange Face Inside Diameter	Fid	2500.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2555.000	mm
Gasket Inside Diameter	Gi	2530.000	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Gasket Contact Width,	N = Self-Energizing	0.000 mm
Basic Gasket Width,	bo = N/2	0.000 mm
Effective Gasket Width,	b = bo	0.000 mm
Gasket Reaction Diameter,	G = Go (Self-Energizing)	2555.000 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2555.0^2 \cdot 163.157$$

$$= 8365192.500 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 0.0 \cdot \pi \cdot 2555.0 \cdot 0.0 \cdot 163.16$$

$$= 0.000 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(8365193 + 0 + 0, 0)$$

$$= 8365192.500 \text{ kgf}$$



Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 0.0 \cdot 0.0 \cdot \pi \cdot 2555.0 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 0.000 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a)$$

$$= \max((8365193 + 0 + 4 \cdot 0 / 2555.0) / 2039, 0 / 2039)$$

$$= 4101.948 \text{ cm}^2$$

Bolting Information for User Defined Bolts (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4101.948	4431.840	
Circ. Spacing between the Bolts:	...	314.732	...

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 2039.32 \cdot 4431.8403$$

$$= 9037949.00 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 8365192.50 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2811.0 - 2555.0) / 2$$

$$= 128.0000 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 8365193 \cdot 128.0 \cdot 1.0 / (537.0^2 \cdot 2555.0) + 0.3 \cdot 1.0 \cdot 163.16 \cdot 2555.0^2 / 537.0^2$$

$$= 1384.17 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot h_G \cdot B_{sc} / (t^2 \cdot G)$$

$$= 1.9 \cdot 9037949 \cdot 128.0 \cdot 1.0 / (537.0^2 \cdot 2555.0)$$

$$= 298.33 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 8365193 / 4431.8403$$

$$= 1887.52 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (0 / 4431.8403)$$

$$= 0.00 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	1384.17	1488.73	298.33	1876.18
Bolting	1887.52	2039.32	0.00	2039.32

Reqd. Blind Flange Thickness at Center 520.799 mm
 Reqd. Blind Flange Thickness at Gasket 234.269 mm

Estimated M.A.W.P. (Operating) 175 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 30980.1 kg
 Estimated Unfinished Weight of Forging at given Thk 30980.1 kg

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	1368.84	1876.18	295.02	1876.18
Bolting	1887.52	2039.32	0.00	2039.32

Minimum Req'd Thickness, ambient, no Corrosion: 461.264 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Bottom head		163.23	72	3	2516	1672.3	1
Shell-1		163.23	132	3	2500	1672.3	2
Shell-2		163.22	132	3	2500	1672.3	3
Inte skirt		...	30	...	2735.2	...	4
Shell-3		163.22	132	3	2500	1672.3	5
Shell-4		163.21	132	3	2500	1672.3	6
Shell-5		163.2	132	3	2500	1672.3	7
Shell-6		163.19	132	3	2500	1672.3	8
Shell-7		163.18	132	3	2500	1672.3	9
Shell-8		163.17	132	3	2500	1672.3	10
Shell flange-1		163.16	1380	3	2500	1488.7	11
Shell flange-2		163.16	630	3	3070	1488.7	12

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Bottom head		163.155	165.5	219.045	67	66.0713	1
Shell-1		163.155	163.792	211.889	132	131.474	2
Shell-2		163.155	163.801	211.889	132	131.466	3
Inte skirt		...	No Calc	No Calc	30	No Calc	4
Shell-3		163.155	163.807	211.889	132	131.461	5
Shell-4		163.155	163.816	211.889	132	131.454	6
Shell-5		163.155	163.825	211.889	132	131.446	7
Shell-6		163.155	163.834	211.889	132	131.438	8
Shell-7		163.155	163.844	211.889	132	131.431	9
Shell-8		163.155	163.853	211.889	132	131.423	10
Shell flange-1		163.155	164.077	176.278	1375	219.38	11
Shell flange-2		163.155	175.479	176.278	540	520.799	12

Minimum 164 176

MAWP: 164 kgf/cm², limited by: Shell-1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom head

Spherical Head Design Information:				
Design Pressure	P	163.234 kgf/cm²	Int. Design Temperature	300 °C
Material		SA-387 11	Ext. Design Temperature	300 °C
Length		0.000 cm	External Pressure Chart	CS-3
			UNS Number	K11789
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	2110.7 kgf/cm²
Inside Diameter	Di	2516.000 mm	Actual Stress	1568.3 kgf/cm²
Spec. Min. Thk t _e , or s _a		67.0000 mm	Spec. Nominal Thk.	72.000 mm
Surface Area		111143.3 cm²	SG of Contents	0.031
Element Volume		4169668.8 cm³	Weight of Contents	128.1 kgf
Empty Weight		5872.4 kgf	Operating Weight	6000.5 kgf
			Comp. All. Ope. Stress	908.4 kgf/cm²

Radiography Information:				
...		...		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.450 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.725 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2(\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 2522.0/2(\exp(0.5 \cdot 163.234/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 63.0713 + 3.0 + 0.0 = 66.0713 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 1672.3 \cdot 1.0 \cdot \ln((2 \cdot 64.0 + 2522.0) / 2522.0) - 0.079$
 $= 165.579 - 0.079 = 165.5001 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 67.0 + 2516.0) / 2516.0)$
 $= 219.0455 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D^2 / (E(D_o^2 - D^2))$
 $= 163.234 \cdot 2522.0^2 / (1.0(2650.0^2 - 2522.0^2))$
 $= 1568.314 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 0.5 \cdot 163.234 / (1.0 \cdot \ln((2 \cdot 64.0 + 2522.0) / 2522.0))$
 $= 1648.585 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 4.173 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$ 40.223 %

Where D_b is the blank diameter for one piece doubly curved surfaces [Db]:

= square root(4 • Head Surface Area / 3.14159)
 = square root(4 • 111143.3/3.14159)
 = 3761.808 mm

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.450 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.725 mm

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Bottom head

Spherical Head From 10 to 20 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= 0.075 • Ey(t / Ro)
 = 0.075 • 1901117(64.0/1325.0)
 = 6887.067 kgf/cm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Sy
 = 6887.1 / 2508.4
 = 2.746

Engineering Yield to Tensile Stress ratio [R]:

= σys / σuts
 = 2508.4/5273.0
 = 0.476

Ratio [AE]:

= Fhe / E
 = 6887.1/1901117
 = 0.003623

Factor [m1]:

= (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))
 = (ln(0.476) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))
 = 0.162

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= σys(1 + epsys) / ln(1 + epsys)m1
 = 2508.4(1 + 0.002) / ln(1 + 0.002) 0.161796
 = 6870.968 kgf/cm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= m2f(m2f2 - R)
 = 0.6(1.0 - 0.476)
 = 0.315

Curve Fitting Constant [A2]:

= (σuts • exp(m2)) / (m2 m2)
 = (5273.0 • exp(0.315)) / (0.315 0.315)
 = 10391.606 kgf/cm²

Material Parameter for Stress-Strain Curve Model [K]:

= 1.5 • R1.5 - 0.5 • R2.5 - R3.5
 = 1.5 • 0.4761.5 - 0.5 • 0.4762.5 - 0.4763.5
 = 0.340

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
--	-----	---	----	----	----	----	--------------------	-------------------



1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.185E-01
2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.270E-02
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.322E-03
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.183E-02
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.995E-03
6	1894	-3.306	0.569E-06	0.565E-06	0.375E-05	-0.372E-05	0.590E+06	0.410E-03
7	1936	-3.218	0.636E-06	0.631E-06	0.393E-05	-0.389E-05	0.544E+06	0.641E-04
8	1956	-3.174	0.672E-06	0.666E-06	0.402E-05	-0.397E-05	0.522E+06	0.124E-03
9	1946	-3.196	0.653E-06	0.648E-06	0.398E-05	-0.393E-05	0.533E+06	0.284E-04
10	1941	-3.207	0.645E-06	0.640E-06	0.395E-05	-0.391E-05	0.539E+06	0.182E-04
11	1943	-3.202	0.649E-06	0.644E-06	0.397E-05	-0.392E-05	0.536E+06	0.506E-05
12	1942	-3.204	0.647E-06	0.642E-06	0.396E-05	-0.391E-05	0.537E+06	0.657E-05
13	1943	-3.203	0.648E-06	0.643E-06	0.396E-05	-0.392E-05	0.537E+06	0.761E-06
14	1943	-3.202	0.648E-06	0.643E-06	0.396E-05	-0.392E-05	0.536E+06	0.215E-05
15	1943	-3.203	0.648E-06	0.643E-06	0.396E-05	-0.392E-05	0.536E+06	0.692E-06
16	1943	-3.203	0.648E-06	0.643E-06	0.396E-05	-0.392E-05	0.536E+06	0.347E-07
17	1943	-3.203	0.648E-06	0.643E-06	0.396E-05	-0.392E-05	0.536E+06	0.329E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1943.39/2508.4$
 $= 1.833$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1943.39/1.833$
 $= 1060.277 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha}[t/R_o]$
 $= 2 \cdot 1060.277[64.0/1325.0]$
 $= 102.427 \text{ kgf/cm}^2$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	163.230 kgf/cm²	Int. Design Temperature	300 °C	
Material		SA-387 11	Ext. Design Temperature	300 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	1672.3 kgf/cm²	
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	2110.7 kgf/cm²	
Inside Diameter	Di	2500.000 mm	Actual Stress	1585.5 kgf/cm²	
Spec. Min. Thk t,e, or sa		132.0000 mm	Spec. Nominal Thk.	132.000 mm	
Surface Area		260500.9 cm²	SG of Contents	0.031	
Element Volume		14726250.0 cm³	Weight of Contents	452.4 kgf	
Empty Weight		25377.8 kgf	Operating Weight	25830.2 kgf	
			Comp. All. Ope. Stress	999.2 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2506.0 \cdot (\exp(163.23/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 128.4737 + 3.0 + 0.0 = 131.4737 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672.27 \cdot 1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0) - 0.075$
 $= 163.867 - 0.075 = 163.7919 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 132.0 + 2500.0) / 2500.0)$
 $= 211.8893 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 163.23 \cdot 2506.0 / (1.0 (2764.0 - 2506.0))$
 $= 1585.4868 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163.23 / (1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0))$
 $= 1665.7701 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.015 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 22819.336 / \sqrt{(1382.0 \cdot 129.0)}$$

$$= 54.045$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 \cdot (2764.0/129.0)$$

$$= 0.0257$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.026 \cdot 1901117(129.0/2764.0)$$

$$= 3644.139 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3644.1/2508.4$$

$$= 1.453$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2508.4/5273.0$$

$$= 0.476$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3644.1/1901117$$

$$= 0.001917$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.476) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2508.4(1 + 0.002) / \ln(1 + 0.002)^{0.161796}$$

$$= 6870.968 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.476)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (5273.0 \cdot \exp(0.315)) / (0.315^{0.315})$$

$$= 10391.606 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4761.5 - 0.5 \cdot 0.4762.5 - 0.4763.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.202E-01



2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.993E-03
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.203E-02
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.129E-03
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.711E-03
6	1730	-3.656	0.355E-06	0.354E-06	0.308E-05	-0.306E-05	0.801E+06	0.243E-03
7	1689	-3.744	0.313E-06	0.313E-06	0.292E-05	-0.291E-05	0.860E+06	0.467E-04
8	1668	-3.788	0.294E-06	0.293E-06	0.284E-05	-0.283E-05	0.891E+06	0.435E-04
9	1678	-3.766	0.304E-06	0.303E-06	0.288E-05	-0.287E-05	0.875E+06	0.981E-06
10	1673	-3.777	0.299E-06	0.298E-06	0.286E-05	-0.285E-05	0.883E+06	0.214E-04
11	1676	-3.772	0.301E-06	0.300E-06	0.287E-05	-0.286E-05	0.879E+06	0.103E-04
12	1677	-3.769	0.302E-06	0.302E-06	0.288E-05	-0.287E-05	0.877E+06	0.465E-05
13	1678	-3.767	0.303E-06	0.302E-06	0.288E-05	-0.287E-05	0.876E+06	0.183E-05
14	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.428E-06
15	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.277E-06
16	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.755E-07
17	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.101E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1678.52/2508.4$
 $= 1.911$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1678.516/1.911$
 $= 878.273 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha} (t / D_o)$
 $= 2 \cdot 878.273 (129.0/2764.0)$
 $= 81.981 \text{ kgf/cm}^2$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:				
Design Pressure	P	163.221 kgf/cm²	Int. Design Temperature	300 °C
Material		SA-387 11	Ext. Design Temperature	300 °C
Length		185.000 cm	External Pressure Chart	CS-3
			UNS Number	K11789
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	2110.7 kgf/cm²
Inside Diameter	Di	2500.000 mm	Actual Stress	1585.4 kgf/cm²
Spec. Min. Thk t,e, or sa		132.0000 mm	Spec. Nominal Thk.	132.000 mm
Surface Area		160642.2 cm²	SG of Contents	0.031
Element Volume		9081188.0 cm³	Weight of Contents	278.7 kgf
Empty Weight		15649.6 kgf	Operating Weight	15928.3 kgf
			Comp. All. Ope. Stress	999.2 kgf/cm²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2506.0 \cdot (\exp(163.221/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 128.4661 + 3.0 + 0.0 = 131.4661 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672.27 \cdot 1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0) - 0.066$
 $= 163.867 - 0.066 = 163.8011 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 132.0 + 2500.0) / 2500.0)$
 $= 211.8893 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 163.221 \cdot 2506.0 / (1.0(2764.0 - 2506.0))$
 $= 1585.3973 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163.221 / (1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0))$
 $= 1665.6761 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.015 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 22819.336 / \sqrt{(1382.0 \cdot 129.0)}$$

$$= 54.045$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 \cdot (2764.0/129.0)$$

$$= 0.0257$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.026 \cdot 1901117(129.0/2764.0)$$

$$= 3644.139 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3644.1/2508.4$$

$$= 1.453$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2508.4/5273.0$$

$$= 0.476$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3644.1/1901117$$

$$= 0.001917$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.476) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2508.4(1 + 0.002) / \ln(1 + 0.002)^{0.161796}$$

$$= 6870.968 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f^2 - R)$$

$$= 0.6(1.0 - 0.476)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (5273.0 \cdot \exp(0.315)) / (0.315^{0.315})$$

$$= 10391.606 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4761.5 - 0.5 \cdot 0.4762.5 - 0.4763.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.202E-01



2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.993E-03
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.203E-02
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.129E-03
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.711E-03
6	1730	-3.656	0.355E-06	0.354E-06	0.308E-05	-0.306E-05	0.801E+06	0.243E-03
7	1689	-3.744	0.313E-06	0.313E-06	0.292E-05	-0.291E-05	0.860E+06	0.467E-04
8	1668	-3.788	0.294E-06	0.293E-06	0.284E-05	-0.283E-05	0.891E+06	0.435E-04
9	1678	-3.766	0.304E-06	0.303E-06	0.288E-05	-0.287E-05	0.875E+06	0.981E-06
10	1673	-3.777	0.299E-06	0.298E-06	0.286E-05	-0.285E-05	0.883E+06	0.214E-04
11	1676	-3.772	0.301E-06	0.300E-06	0.287E-05	-0.286E-05	0.879E+06	0.103E-04
12	1677	-3.769	0.302E-06	0.302E-06	0.288E-05	-0.287E-05	0.877E+06	0.465E-05
13	1678	-3.767	0.303E-06	0.302E-06	0.288E-05	-0.287E-05	0.876E+06	0.183E-05
14	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.428E-06
15	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.277E-06
16	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.755E-07
17	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.101E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1678.52/2508.4$
 $= 1.911$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1678.516/1.911$
 $= 878.273 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha} (t / D_o)$
 $= 2 \cdot 878.273 (129.0/2764.0)$
 $= 81.981 \text{ kgf/cm}^2$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	163.216 kgf/cm²	Int. Design Temperature	300 °C	
Material		SA-387 11	Ext. Design Temperature	300 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2500.000 mm	Actual Stress	1585.3 kgf/cm²	
Spec. Min. Thk t,e, or sa		132.0000 mm	Spec. Nominal Thk.	132.000 mm	
Surface Area		260500.9 cm²	SG of Contents	0.031	
Element Volume		14726250.0 cm³	Weight of Contents	451.9 kgf	
Empty Weight		25377.8 kgf	Operating Weight	25829.7 kgf	
			Comp. All. Ope. Stress	999.2 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2506.0 \cdot (\exp(163.216/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 128.4613 + 3.0 + 0.0 = 131.4613 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672.27 \cdot 1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0) - 0.061$
 $= 163.867 - 0.061 = 163.8068 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 132.0 + 2500.0) / 2500.0)$
 $= 211.8893 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 163.216 \cdot 2506.0 / (1.0(2764.0 - 2506.0))$
 $= 1585.3423 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163.216 / (1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0))$
 $= 1665.6182 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.015 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 22819.336 / \sqrt{(1382.0 \cdot 129.0)}$$

$$= 54.045$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 \cdot (2764.0/129.0)$$

$$= 0.0257$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.026 \cdot 1901117(129.0/2764.0)$$

$$= 3644.139 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3644.1/2508.4$$

$$= 1.453$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2508.4/5273.0$$

$$= 0.476$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3644.1/1901117$$

$$= 0.001917$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.476) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2508.4(1 + 0.002) / \ln(1 + 0.002)^{0.161796}$$

$$= 6870.968 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f^2 - R)$$

$$= 0.6(1.0 - 0.476)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (5273.0 \cdot \exp(0.315)) / (0.315^{0.315})$$

$$= 10391.606 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4761.5 - 0.5 \cdot 0.4762.5 - 0.4763.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.202E-01



2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.993E-03
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.203E-02
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.129E-03
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.711E-03
6	1730	-3.656	0.355E-06	0.354E-06	0.308E-05	-0.306E-05	0.801E+06	0.243E-03
7	1689	-3.744	0.313E-06	0.313E-06	0.292E-05	-0.291E-05	0.860E+06	0.467E-04
8	1668	-3.788	0.294E-06	0.293E-06	0.284E-05	-0.283E-05	0.891E+06	0.435E-04
9	1678	-3.766	0.304E-06	0.303E-06	0.288E-05	-0.287E-05	0.875E+06	0.981E-06
10	1673	-3.777	0.299E-06	0.298E-06	0.286E-05	-0.285E-05	0.883E+06	0.214E-04
11	1676	-3.772	0.301E-06	0.300E-06	0.287E-05	-0.286E-05	0.879E+06	0.103E-04
12	1677	-3.769	0.302E-06	0.302E-06	0.288E-05	-0.287E-05	0.877E+06	0.465E-05
13	1678	-3.767	0.303E-06	0.302E-06	0.288E-05	-0.287E-05	0.876E+06	0.183E-05
14	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.428E-06
15	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.277E-06
16	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.755E-07
17	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.101E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1678.52/2508.4$
 $= 1.911$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1678.516/1.911$
 $= 878.273 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha} (t / D_o)$
 $= 2 \cdot 878.273 (129.0/2764.0)$
 $= 81.981 \text{ kgf/cm}^2$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	163.206 kgf/cm²	Int. Design Temperature	300 °C	
Material		SA-387 11	Ext. Design Temperature	300 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2500.000 mm	Actual Stress	1585.3 kgf/cm²	
Spec. Min. Thk t,e, or sa		132.0000 mm	Spec. Nominal Thk.	132.000 mm	
Surface Area		260500.9 cm²	SG of Contents	0.031	
Element Volume		14726250.0 cm³	Weight of Contents	452.4 kgf	
Empty Weight		25377.8 kgf	Operating Weight	25830.2 kgf	
			Comp. All. Ope. Stress	999.2 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2506.0 \cdot (\exp(163.206/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 128.4537 + 3.0 + 0.0 = 131.4537 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672.27 \cdot 1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0) - 0.051$
 $= 163.867 - 0.051 = 163.816 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 132.0 + 2500.0) / 2500.0)$
 $= 211.8893 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 163.206 \cdot 2506.0 / (1.0(2764.0 - 2506.0))$
 $= 1585.2529 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163.206 / (1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0))$
 $= 1665.5243 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.015 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 22819.336 / \sqrt{(1382.0 \cdot 129.0)}$$

$$= 54.045$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 \cdot (2764.0/129.0)$$

$$= 0.0257$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.026 \cdot 1901117(129.0/2764.0)$$

$$= 3644.139 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3644.1/2508.4$$

$$= 1.453$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_y / \sigma_{ts}$$

$$= 2508.4/5273.0$$

$$= 0.476$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3644.1/1901117$$

$$= 0.001917$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.476) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_y(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2508.4(1 + 0.002) / \ln(1 + 0.002)^{0.161796}$$

$$= 6870.968 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f^2 - R)$$

$$= 0.6(1.0 - 0.476)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{ts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (5273.0 \cdot \exp(0.315)) / (0.315^{0.315})$$

$$= 10391.606 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4761.5 - 0.5 \cdot 0.4762.5 - 0.4763.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.202E-01



2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.993E-03
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.203E-02
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.129E-03
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.711E-03
6	1730	-3.656	0.355E-06	0.354E-06	0.308E-05	-0.306E-05	0.801E+06	0.243E-03
7	1689	-3.744	0.313E-06	0.313E-06	0.292E-05	-0.291E-05	0.860E+06	0.467E-04
8	1668	-3.788	0.294E-06	0.293E-06	0.284E-05	-0.283E-05	0.891E+06	0.435E-04
9	1678	-3.766	0.304E-06	0.303E-06	0.288E-05	-0.287E-05	0.875E+06	0.981E-06
10	1673	-3.777	0.299E-06	0.298E-06	0.286E-05	-0.285E-05	0.883E+06	0.214E-04
11	1676	-3.772	0.301E-06	0.300E-06	0.287E-05	-0.286E-05	0.879E+06	0.103E-04
12	1677	-3.769	0.302E-06	0.302E-06	0.288E-05	-0.287E-05	0.877E+06	0.465E-05
13	1678	-3.767	0.303E-06	0.302E-06	0.288E-05	-0.287E-05	0.876E+06	0.183E-05
14	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.428E-06
15	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.277E-06
16	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.755E-07
17	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.101E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1678.52/2508.4$
 $= 1.911$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1678.516/1.911$
 $= 878.273 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha} (t / D_o)$
 $= 2 \cdot 878.273 (129.0/2764.0)$
 $= 81.981 \text{ kgf/cm}^2$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:				
Design Pressure	P	163.197 kgf/cm²	Int. Design Temperature	300 °C
Material		SA-387 11	Ext. Design Temperature	300 °C
Length		300.000 cm	External Pressure Chart	CS-3
			UNS Number	K11789
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	2110.7 kgf/cm²
Inside Diameter	Di	2500.000 mm	Actual Stress	1585.2 kgf/cm²
Spec. Min. Thk t,e, or sa		132.0000 mm	Spec. Nominal Thk.	132.000 mm
Surface Area		260500.9 cm²	SG of Contents	0.031
Element Volume		14726250.0 cm³	Weight of Contents	453.6 kgf
Empty Weight		25377.8 kgf	Operating Weight	25831.3 kgf
			Comp. All. Ope. Stress	999.2 kgf/cm²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2506.0 \cdot (\exp(163.197/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 128.446 + 3.0 + 0.0 = 131.446 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672.27 \cdot 1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0) - 0.042$
 $= 163.867 - 0.042 = 163.8252 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 132.0 + 2500.0) / 2500.0)$
 $= 211.8893 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 163.197 \cdot 2506.0 / (1.0(2764.0 - 2506.0))$
 $= 1585.1633 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163.197 / (1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0))$
 $= 1665.4302 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.015 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 22819.336 / \sqrt{(1382.0 \cdot 129.0)}$$

$$= 54.045$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 \cdot (2764.0/129.0)$$

$$= 0.0257$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.026 \cdot 1901117(129.0/2764.0)$$

$$= 3644.139 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3644.1/2508.4$$

$$= 1.453$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_y / \sigma_{uts}$$

$$= 2508.4/5273.0$$

$$= 0.476$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3644.1/1901117$$

$$= 0.001917$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / (\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.476) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_y(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2508.4(1 + 0.002) / \ln(1 + 0.002)^{0.161796}$$

$$= 6870.968 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f^2 - R)$$

$$= 0.6(1.0 - 0.476)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (5273.0 \cdot \exp(0.315)) / (0.315^{0.315})$$

$$= 10391.606 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4761.5 - 0.5 \cdot 0.4762.5 - 0.4763.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.202E-01



2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.993E-03
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.203E-02
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.129E-03
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.711E-03
6	1730	-3.656	0.355E-06	0.354E-06	0.308E-05	-0.306E-05	0.801E+06	0.243E-03
7	1689	-3.744	0.313E-06	0.313E-06	0.292E-05	-0.291E-05	0.860E+06	0.467E-04
8	1668	-3.788	0.294E-06	0.293E-06	0.284E-05	-0.283E-05	0.891E+06	0.435E-04
9	1678	-3.766	0.304E-06	0.303E-06	0.288E-05	-0.287E-05	0.875E+06	0.981E-06
10	1673	-3.777	0.299E-06	0.298E-06	0.286E-05	-0.285E-05	0.883E+06	0.214E-04
11	1676	-3.772	0.301E-06	0.300E-06	0.287E-05	-0.286E-05	0.879E+06	0.103E-04
12	1677	-3.769	0.302E-06	0.302E-06	0.288E-05	-0.287E-05	0.877E+06	0.465E-05
13	1678	-3.767	0.303E-06	0.302E-06	0.288E-05	-0.287E-05	0.876E+06	0.183E-05
14	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.428E-06
15	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.277E-06
16	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.755E-07
17	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.101E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1678.52/2508.4$
 $= 1.911$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1678.516/1.911$
 $= 878.273 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha} (t / D_o)$
 $= 2 \cdot 878.273 (129.0/2764.0)$
 $= 81.981 \text{ kgf/cm}^2$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	163.188 kgf/cm²	Int. Design Temperature		300 °C
Material		SA-387 11	Ext. Design Temperature		300 °C
Length		300.000 cm	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2500.000 mm	Actual Stress		1585.1 kgf/cm²
Spec. Min. Thk t,e, or sa		132.0000 mm	Spec. Nominal Thk.		132.000 mm
Surface Area		260500.9 cm²	SG of Contents		0.031
Element Volume		14726250.0 cm³	Weight of Contents		453.4 kgf
Empty Weight		25377.8 kgf	Operating Weight		25831.1 kgf
			Comp. All. Ope. Stress		999.2 kgf/cm²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2506.0 \cdot (\exp(163.188/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 128.4384 + 3.0 + 0.0 = 131.4384 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672.27 \cdot 1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0) - 0.033$
 $= 163.867 - 0.033 = 163.8345 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 132.0 + 2500.0) / 2500.0)$
 $= 211.8893 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 163.188 \cdot 2506.0 / (1.0(2764.0 - 2506.0))$
 $= 1585.0735 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163.188 / (1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0))$
 $= 1665.3359 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.015 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-6

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 22819.332 / \sqrt{(1382.0 \cdot 129.0)}$$

$$= 54.045$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (2764.0/129.0)$$

$$= 0.0257$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.026 \cdot 1901117(129.0/2764.0)$$

$$= 3644.139 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3644.1/2508.4$$

$$= 1.453$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2508.4/5273.0$$

$$= 0.476$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3644.1/1901117$$

$$= 0.001917$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.476) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2508.4(1 + 0.002) / \ln(1 + 0.002)^{0.161796}$$

$$= 6870.968 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.476)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (5273.0 \cdot \exp(0.315)) / (0.315^{0.315})$$

$$= 10391.606 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4761.5 - 0.5 \cdot 0.4762.5 - 0.4763.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.202E-01



2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.993E-03
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.203E-02
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.129E-03
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.711E-03
6	1730	-3.656	0.355E-06	0.354E-06	0.308E-05	-0.306E-05	0.801E+06	0.243E-03
7	1689	-3.744	0.313E-06	0.313E-06	0.292E-05	-0.291E-05	0.860E+06	0.467E-04
8	1668	-3.788	0.294E-06	0.293E-06	0.284E-05	-0.283E-05	0.891E+06	0.435E-04
9	1678	-3.766	0.304E-06	0.303E-06	0.288E-05	-0.287E-05	0.875E+06	0.981E-06
10	1673	-3.777	0.299E-06	0.298E-06	0.286E-05	-0.285E-05	0.883E+06	0.214E-04
11	1676	-3.772	0.301E-06	0.300E-06	0.287E-05	-0.286E-05	0.879E+06	0.103E-04
12	1677	-3.769	0.302E-06	0.302E-06	0.288E-05	-0.287E-05	0.877E+06	0.465E-05
13	1678	-3.767	0.303E-06	0.302E-06	0.288E-05	-0.287E-05	0.876E+06	0.183E-05
14	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.428E-06
15	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.277E-06
16	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.755E-07
17	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.101E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1678.52/2508.4$
 $= 1.911$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1678.516/1.911$
 $= 878.273 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha} (t / D_o)$
 $= 2 \cdot 878.273 (129.0/2764.0)$
 $= 81.981 \text{ kgf/cm}^2$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	163.179 kgf/cm²	Int. Design Temperature	300 °C	
Material		SA-387 11	Ext. Design Temperature	300 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2500.000 mm	Actual Stress	1585.0 kgf/cm²	
Spec. Min. Thk t,e, or sa		132.0000 mm	Spec. Nominal Thk.	132.000 mm	
Surface Area		260500.9 cm²	SG of Contents	0.031	
Element Volume		14726250.0 cm³	Weight of Contents	453.4 kgf	
Empty Weight		25377.8 kgf	Operating Weight	25831.1 kgf	
			Comp. All. Ope. Stress	999.2 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2506.0 \cdot (\exp(163.179/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 128.4308 + 3.0 + 0.0 = 131.4308 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672.27 \cdot 1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0) - 0.024$
 $= 163.867 - 0.024 = 163.8437 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 132.0 + 2500.0) / 2500.0)$
 $= 211.8893 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 163.179 \cdot 2506.0 / (1.0(2764.0 - 2506.0))$
 $= 1584.9838 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163.179 / (1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0))$
 $= 1665.2415 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.015 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-7

Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 22819.332 / \sqrt{(1382.0 \cdot 129.0)}$$

$$= 54.045$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (2764.0/129.0)$$

$$= 0.0257$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.026 \cdot 1901117(129.0/2764.0)$$

$$= 3644.139 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3644.1/2508.4$$

$$= 1.453$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2508.4/5273.0$$

$$= 0.476$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3644.1/1901117$$

$$= 0.001917$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.476) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2508.4(1 + 0.002) / \ln(1 + 0.002)^{0.161796}$$

$$= 6870.968 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.476)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (5273.0 \cdot \exp(0.315)) / (0.315^{0.315})$$

$$= 10391.606 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4761.5 - 0.5 \cdot 0.4762.5 - 0.4763.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.202E-01



2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.993E-03
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.203E-02
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.129E-03
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.711E-03
6	1730	-3.656	0.355E-06	0.354E-06	0.308E-05	-0.306E-05	0.801E+06	0.243E-03
7	1689	-3.744	0.313E-06	0.313E-06	0.292E-05	-0.291E-05	0.860E+06	0.467E-04
8	1668	-3.788	0.294E-06	0.293E-06	0.284E-05	-0.283E-05	0.891E+06	0.435E-04
9	1678	-3.766	0.304E-06	0.303E-06	0.288E-05	-0.287E-05	0.875E+06	0.981E-06
10	1673	-3.777	0.299E-06	0.298E-06	0.286E-05	-0.285E-05	0.883E+06	0.214E-04
11	1676	-3.772	0.301E-06	0.300E-06	0.287E-05	-0.286E-05	0.879E+06	0.103E-04
12	1677	-3.769	0.302E-06	0.302E-06	0.288E-05	-0.287E-05	0.877E+06	0.465E-05
13	1678	-3.767	0.303E-06	0.302E-06	0.288E-05	-0.287E-05	0.876E+06	0.183E-05
14	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.428E-06
15	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.277E-06
16	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.755E-07
17	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.101E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1678.52/2508.4$
 $= 1.911$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1678.516/1.911$
 $= 878.273 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha} (t / D_o)$
 $= 2 \cdot 878.273 (129.0/2764.0)$
 $= 81.981 \text{ kgf/cm}^2$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-8

Cylindrical Shell Design Information:				
Design Pressure	P	163.169 kgf/cm²	Int. Design Temperature	300 °C
Material		SA-387 11	Ext. Design Temperature	300 °C
Length		255.000 cm	External Pressure Chart	CS-3
			UNS Number	K11789
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	2110.7 kgf/cm²
Inside Diameter	Di	2500.000 mm	Actual Stress	1584.9 kgf/cm²
Spec. Min. Thk t,e, or sa		132.0000 mm	Spec. Nominal Thk.	132.000 mm
Surface Area		221425.8 cm²	SG of Contents	0.031
Element Volume		12517313.0 cm³	Weight of Contents	385.4 kgf
Empty Weight		21571.1 kgf	Operating Weight	21956.5 kgf
			Comp. All. Ope. Stress	999.2 kgf/cm²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2506.0 \cdot (\exp(163.169/(1672.27 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 128.4232 + 3.0 + 0.0 = 131.4232 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672.27 \cdot 1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0) - 0.014$
 $= 163.867 - 0.014 = 163.853 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2110.7 \cdot 1.0 \cdot \ln((2 \cdot 132.0 + 2500.0) / 2500.0)$
 $= 211.8893 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 163.169 \cdot 2506.0 / (1.0(2764.0 - 2506.0))$
 $= 1584.8940 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163.169 / (1.0 \cdot \ln((2 \cdot 129.0 + 2506.0) / 2506.0))$
 $= 1665.1473 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.015 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-8

Cylindrical Shell From 100 to 110 Ext. Chart: CS-2 at 300 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 22819.332 / \sqrt{(1382.0 \cdot 129.0)}$$

$$= 54.045$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (2764.0/129.0)$$

$$= 0.0257$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.026 \cdot 1901117(129.0/2764.0)$$

$$= 3644.139 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3644.1/2508.4$$

$$= 1.453$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2508.4/5273.0$$

$$= 0.476$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3644.1/1901117$$

$$= 0.001917$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.476) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2508.4(1 + 0.002) / \ln(1 + 0.002)^{0.161796}$$

$$= 6870.968 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.476)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (5273.0 \cdot \exp(0.315)) / (0.315^{0.315})$$

$$= 10391.606 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4761.5 - 0.5 \cdot 0.4762.5 - 0.4763.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.727	0.315E-05	0.261E-05	0.770E-05	-0.562E-05	0.119E+06	0.202E-01



2	1318	-4.533	0.868E-07	0.867E-07	0.170E-05	-0.170E-05	0.143E+07	0.993E-03
3	1977	-3.130	0.709E-06	0.703E-06	0.412E-05	-0.406E-05	0.501E+06	0.203E-02
4	1647	-3.832	0.276E-06	0.275E-06	0.277E-05	-0.276E-05	0.921E+06	0.129E-03
5	1812	-3.481	0.452E-06	0.450E-06	0.341E-05	-0.338E-05	0.690E+06	0.711E-03
6	1730	-3.656	0.355E-06	0.354E-06	0.308E-05	-0.306E-05	0.801E+06	0.243E-03
7	1689	-3.744	0.313E-06	0.313E-06	0.292E-05	-0.291E-05	0.860E+06	0.467E-04
8	1668	-3.788	0.294E-06	0.293E-06	0.284E-05	-0.283E-05	0.891E+06	0.435E-04
9	1678	-3.766	0.304E-06	0.303E-06	0.288E-05	-0.287E-05	0.875E+06	0.981E-06
10	1673	-3.777	0.299E-06	0.298E-06	0.286E-05	-0.285E-05	0.883E+06	0.214E-04
11	1676	-3.772	0.301E-06	0.300E-06	0.287E-05	-0.286E-05	0.879E+06	0.103E-04
12	1677	-3.769	0.302E-06	0.302E-06	0.288E-05	-0.287E-05	0.877E+06	0.465E-05
13	1678	-3.767	0.303E-06	0.302E-06	0.288E-05	-0.287E-05	0.876E+06	0.183E-05
14	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.428E-06
15	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.277E-06
16	1678	-3.767	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.755E-07
17	1678	-3.766	0.303E-06	0.303E-06	0.288E-05	-0.287E-05	0.876E+06	0.101E-06

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1678.52/2508.4$
 $= 1.911$

Allowable Circumferential Compressive Stress [Fha]:
 $= F_{ic}/FS$
 $= 1678.516/1.911$
 $= 878.273 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot F_{ha} (t / D_o)$
 $= 2 \cdot 878.273 (129.0/2764.0)$
 $= 81.981 \text{ kgf/cm}^2$

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Hydrostatic Test Pressure Results:

User Defined Hydrostatic Test Pressure at High Point 260.871 kgf/cm²

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	2516.2	3002.9	0.838	261.12
Shell-1	2601.1	3002.9	0.866	261.12
Shell-2	2601.1	3002.9	0.866	261.12
Shell-3	2601.1	3002.9	0.866	261.12
Shell-4	2601.1	3002.9	0.866	261.12
Shell-5	2601.1	3002.9	0.866	261.12
Shell-6	2601.1	3002.9	0.866	261.12
Shell-7	2601.1	3002.9	0.866	261.12
Shell-8	2601.1	3002.9	0.866	261.12

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B 450mm	Nozzle	2110.70	1723.24	1.225
Top plug weight	Nozzle	2110.74	1723.26	1.225
AT 2"	Nozzle	2110.74	1723.26	1.225
BT 2"	Nozzle	2110.74	1723.26	1.225

Minimum 1.225

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom head	2110.70	1672.27	1.262
Shell-1	2110.70	1672.27	1.262
Shell-2	2110.70	1672.27	1.262
Shell-3	2110.70	1672.27	1.262
Shell-4	2110.70	1672.27	1.262
Shell-5	2110.70	1672.27	1.262
Shell-6	2110.70	1672.27	1.262
Shell-7	2110.70	1672.27	1.262
Shell-8	2110.70	1672.27	1.262
Shell flange-1	1876.18	1488.73	1.260
Shell flange-2	1876.18	1488.73	1.260

Minimum 1.260

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
B 450mm	1759.35	3002.91	0.586
Top plug weight	716.26	3002.96	0.239
AT 2"	953.88	3002.96	0.318
BT 2"	953.88	3002.96	0.318

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	2650	64	No Calc	No Calc
20	30	2281.93	2764	129	No Calc	No Calc
30	40	2281.93	2764	129	No Calc	No Calc
40	50	No Calc	...	...	No Calc	No Calc
50	60	2281.93	2764	129	No Calc	No Calc
60	70	2281.93	2764	129	No Calc	No Calc
70	80	2281.93	2764	129	No Calc	No Calc
80	90	2281.93	2764	129	No Calc	No Calc
90	100	2281.93	2764	129	No Calc	No Calc
100	110	2281.93	2764	129	No Calc	No Calc
110	120	No Calc	...	1372	No Calc	No Calc
120	130	No Calc	...	537	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	Design External Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	67	6.60446	1.05456	102.427
20	30	132	23.6136	1.05456	81.9807
30	40	132	23.6136	1.05456	81.9807
40	50	...	No Calc	...	No Calc
50	60	132	23.6136	1.05456	81.9807
60	70	132	23.6136	1.05456	81.9807
70	80	132	23.6136	1.05456	81.9807
80	90	132	23.6136	1.05456	81.9807
90	100	132	23.6136	1.05456	81.9807
100	110	132	23.6136	1.05456	81.9807
110	120	1375	219.38	1.05456	No Calc
120	130	540	217.145	1.05456	No Calc

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2281.93	2539997	No Calc	No Calc
30	40	2281.93	2539997	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	2281.93	2539997	No Calc	No Calc
60	70	2281.93	2539997	No Calc	No Calc
70	80	2281.93	2539997	No Calc	No Calc
80	90	2281.93	2539997	No Calc	No Calc
90	100	2281.93	2539997	No Calc	No Calc
100	110	2281.93	2539997	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	5872.39	4169668	5640.64	4199570	195.551
20	30	25377.8	14726250	24829.3	14797021	845.079
30	40	15649.6	9081188	15311.4	9124830	521.132
40	50	8390.7	...	8390.7	...	279.41
50	60	25377.8	14726250	24829.3	14797021	845.079
60	70	25377.8	14726250	24829.3	14797021	845.079
70	80	25377.8	14726250	24829.3	14797021	845.079
80	90	25377.8	14726250	24829.3	14797021	845.079
90	100	25377.8	14726250	24829.3	14797021	845.079
100	110	21571.1	12517313	21104.9	12577469	718.318
110	120	29156.8	7911099	28863.4	7916406	970.922
120	130	36143.4	...	36143.4	...	1203.58
Total		269050	0.122E+09	264429	0.123E+09	8959

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	128.093	...	-62.9	...	Liquid: 10
10	Insl	218.548	...	-62.9	...	Ins: 90
10	Nozl	1589.29	...	-162.45	...	B 450mm
10	Wght	800	130	...	...	SF weight 1
10	Wght	800	-130	...	...	SF weight 2
20	Liqd	452.394	...	150	...	Liquid: 20
20	Insl	489.823	...	150	...	Ins: 90
20	Wght	2999.95	146	...	...	Support ring1
20	Wght	2999.95	-146	...	...	Support ring2
20	Wght	3499.94	185	220	...	Tailing lug
30	Liqd	278.666	...	92.5	...	Liquid: 30
30	Insl	302.058	...	92.5	...	Ins: 90
30	Wght	97750	...	185	...	Catalyst-3
30	Wght	3000	140	185	...	Skir band-1
30	Wght	3000	-140	185	...	Skirt band-2
30	Wght	25365	...	185	...	Basket-3
50	Liqd	451.891	...	150	...	Liquid: 50
50	Insl	489.823	...	150	...	Ins: 90
60	Liqd	452.394	...	150	...	Liquid: 60
60	Insl	489.823	...	150	...	Ins: 90
70	Liqd	453.572	...	150	...	Liquid: 70
70	Insl	489.823	...	150	...	Ins: 90
80	Liqd	453.364	...	150	...	Liquid: 80
80	Insl	489.823	...	150	...	Ins: 90
80	Wght	41950	...	149.997	...	Catalyst-2
80	Wght	25365	...	150	...	Basket-2
90	Liqd	453.364	...	150	...	Liquid: 90
90	Insl	489.823	...	150	...	Ins: 90
100	Liqd	385.36	...	127.5	...	Liquid: 100
100	Insl	416.35	...	127.5	...	Ins: 90
100	Wght	2251.96	...	99.9974	...	Cleat weight
100	Wght	25750	...	100	...	Catalyst-1
100	Wght	25365	...	100	...	Basket-1
100	Wght	3500	138.2	145	...	Trunnion-1
100	Wght	3500	-138.2	145	...	Trunnion-2
100	Wght	8455	...	255	...	Elec heater
100	Wght	850	135	255	...	Nozzle-A
100	Wght	270	-135	255	...	Nozzle-C
100	Wght	270	...	...	135	Nozzle-D
110	Liqd	241.277	...	80	...	Liquid: 110
110	Insl	70.602	...	38.25	...	Ins: 90
120	Liqd	...	...	27	...	Liquid: 120



120	Insl	190.21	...	27	...
120	Nozl	1673.95	...	55.2497	...
120	Nozl	65.45	86.6025	104	-50
120	Nozl	65.45	-86.6025	104	50

Ins: 90
 Top plug weight
 AT 2"
 BT 2"

Total Weight of Each Detail Type:

Liquid	3750.4
Insulation	4136.7
Nozzles	3394.1
Weights	277741.8
Sum of the Detail Weights	289023.0 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	278010.0	278010.0	278010.0	278010.0	278010.0	278010.0
Nozzles	3394.1	3394.1	3394.1	3394.1	3394.1	3394.1
Wld Weights	27741.8	27741.8	27741.8	27741.8	27741.8	27741.8
Insulation	...	...	...	4136.7	4136.7	4136.7
Empty Weights	...	...	...	250000.0	250000.0	...
Ope Weights	...	...	...	...	...	250000.0
Ope. Liquid	...	...	...	...	...	3750.4
Test Liquid	...	121981.7	...	...	...	...
Totals	309146.0	431127.7	309146.0	563282.7	563282.7	567033.1

Field Installation Options:

Insulation installed before lifting.

Miscellaneous Weight Percent: 3.3 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	309146.0 kg
Shop Test Wt.	- Fabricated weight + water (full)	431127.7 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	309146.0 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	563282.7 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	563282.7 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	567033.1 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	193224.8 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	167223.5 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	193515.0 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	111143	99435.4
20	30	260501	235619
30	40	160642	145299
40	50	263442	257787
50	60	260501	235619
60	70	260501	235619
70	80	260501	235619
80	90	260501	235619
90	100	260501	235619
100	110	221426	200277



110	120	178413	126056
120	130	134785	...

Total 2632856.2 2242571.2 cm²
 Total 2834.0 2413.9 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	9257.23	9603.87	13643.6	...	9475.78
20	30	35722.7	36664.9	50932.1	6474.89	36212.5
30	40	145286	145866	31549.9	...	145588
40	50	8670.11	8670.11	8670.11	...	8670.11
50	60	26222.8	27164.6	41432.3	...	26712.7
60	70	26222.8	27165.1	41432.3	...	26712.7
70	80	26222.8	27166.2	41432.3	...	26712.7
80	90	93537.8	94481	41432.3	...	94027.7
90	100	26222.8	27166	41432.3	...	26712.7
100	110	92501.4	93303.1	45859.4	783.001	92917.7
110	120	30127.7	30439.6	38105.9	...	30198.3
120	130	39151.9	39342.1	39342.1	...	39342.1

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	...	...	...
20	30	-9475.82	-9603.85	-13643.6
30	40	-45688.3	-46268.8	-64575.7
40	50	563283	567033	339139
50	60	363336	366228	330469
60	70	336624	339063	289036
70	80	309911	311898	247604
80	90	283198	284732	206172
90	100	189171	190251	164740
100	110	162458	163085	123307
110	120	69540.4	69781.7	77447.9
120	130	39342.1	39342.1	39342.1

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	...	...	...
20	30	-6474.89	-6474.89	-6474.89
30	40	-6474.89	-6474.89	-6474.89
40	50	783.001	783.001	783.001
50	60	783.001	783.001	783.001
60	70	783.001	783.001	783.001
70	80	783.001	783.001	783.001
80	90	783.001	783.001	783.001
90	100	783.001	783.001	783.001
100	110	783.001	783.001	783.001
110	120	...	...	...
120	130	...	...	...





Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
B 450mm	229.54	263.70	300	1500	GR 1.10	...	...	...	...
AT 2"	229.54	263.70	300	1500	GR 1.9	...	...	...	...
BT 2"	229.54	263.70	300	1500	GR 1.9	...	...	...	...
Min Rating	229.545	263.705, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 2.14721 Hz.

The Natural Frequency for the Vessel (Ope...) is 2.14167 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	900	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class B
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 7.283

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 2.142 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (7.283) is > 5 .

Note:

As per Section 9.1, Dynamic effects of wind should be considered. The Cross-Wind dynamic effects can be considered using the Vortex Shedding which can be invoked from Design Constraints tab. While the Along-Wind effects can be considered using Gust Response Factor method, which can be invoked from the Wind Load tab.

Design Wind Speed [Vz]:

$$= \text{Basic Wind Speed} \cdot k_1 \cdot k_{2,1} \cdot k_3 \cdot k_4 \quad (\text{for the 1st element})$$

$$= 50.0 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 1.15$$

$$= 57.496 \text{ meters/s}$$

Design Wind Pressure per 4.5.2 [pd]:

$$= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$$

$$= \max(0.7 \cdot 1983.503, 1.0 \cdot 0.8 \cdot 1.0 \cdot 0.6 \cdot 57.5^2)$$

$$= 1586.8 \text{ N/m}^2$$

Element Wind Load (operating condition):

$$= \text{Wind Area} \cdot C_f \cdot p_d$$

$$= 4.4 \cdot 0.846 \cdot 1586.803$$

$$= 5853.8 \text{ N}$$

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	9.83	1.0000	1.0000	1.0000	57.50	0.8457
20	11.76	1.0000	1.0176	1.0000	58.51	0.8457
30	14.18	1.0000	1.0418	1.0000	59.90	0.8457
40	13.53	1.0000	1.0353	1.0000	59.53	0.8457
50	16.61	1.0000	1.0564	1.0000	60.74	0.8457
60	19.61	1.0000	1.0684	1.0000	61.43	0.8457
70	22.61	1.0000	1.0830	1.0000	62.27	0.8457
80	25.61	1.0000	1.0980	1.0000	63.13	0.8457
90	28.61	1.0000	1.1130	1.0000	64.00	0.8457
100	31.38	1.0000	1.1235	1.0000	64.59	0.8457
110	33.46	1.0000	1.1287	1.0000	64.89	0.8457

120 34.53 1.0000 1.1313 1.0000 65.05 0.8457

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, cm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, cm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure cm
 Lc - Total length of conical section(s) of vessel, cm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
 Z - Maximum amplitude of vibration at top of vessel, mm
 Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, ≤ 0.00032 (High); $0.00032 < 0.0004$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/2454.318 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[11.11/80.52^2] \cdot 10^4 = 17.136$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.0004$ no chance. [Vp]:

= $W / (L \cdot Dr^2)$

= $562259 / (2454.32 \cdot 308.158^2)$

= $0.24125E-02$

Since V_p is > 0.0004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
SF weight 1	0.00	900.00	161.88	0.00
SF weight 2	0.00	900.00	161.88	0.00
Support ring1	0.00	900.00	161.88	0.00
Support ring2	0.00	900.00	161.88	0.00
Tailing lug	0.00	1120.00	165.95	0.00
Catalyst-3	0.00	1385.00	174.59	0.00
Skir band-1	0.00	1385.00	174.59	0.00
Skirt band-2	0.00	1385.00	174.59	0.00
Basket-3	0.00	1385.00	174.59	0.00
Catalyst-2	0.00	2435.00	192.95	0.00
Basket-2	0.00	2435.00	192.95	0.00
Cleat weight	0.00	2985.00	202.23	0.00
Catalyst-1	0.00	2985.00	202.23	0.00
Basket-1	0.00	2985.00	202.23	0.00
Trunnion-1	0.00	3030.00	202.84	0.00
Trunnion-2	0.00	3030.00	202.84	0.00
Elec heater	0.00	3140.00	204.33	0.00
Nozzle-A	0.00	3140.00	204.33	0.00
Nozzle-C	0.00	3140.00	204.33	0.00
Nozzle-D	0.00	2885.00	200.87	0.00

The Natural Frequency for the Vessel (Ope...) is 2.14167 Hz.

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	983.174	384.37	43623.3	161.875	656.643
20	30	1175.8	399.532	119860	167.617	1868.19
30	40	1418.3	399.532	73913.4	175.701	1207.61
40	50	1353.21	436.866	131060	173.512	2114.61
50	60	1660.8	399.532	119860	180.661	2013.57
60	70	1960.8	399.532	119860	184.788	2059.57
70	80	2260.8	399.532	119860	189.876	2116.28
80	90	2560.8	399.532	119860	195.172	2175.3
90	100	2860.8	399.532	119860	200.54	2235.14
100	110	3138.3	399.532	101881	204.312	1935.6
110	120	3346.12	347.762	55641.9	206.206	1066.92
120	130	3453.12	440.23	23772.4	207.185	457.995

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Input Values:

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1040	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Computed Natural Frequencies (OPE): 8

Mode	Freq (Hz)	Freq (Rad/Sec)	Period (Sec)
1	2.1417	13.4565	0.4669
2	15.9519	100.2288	0.0627
3	25.8734	162.5674	0.0386
4	44.6389	280.4743	0.0224
5	44.8606	281.8676	0.0223
6	73.7566	463.4263	0.0136
7	99.0817	622.5488	0.0101
8	101.3930	637.0711	0.0099

Mass Participation Factors: 8

Mode	X	Y
1	13.49201463	0.00000000
2	0.08465563	-0.00000000
3	-0.00000000	0.04591815
4	0.00832380	-0.00000000
5	0.00571723	-0.00000000
6	0.00055663	-0.00000000
7	-0.00086598	-0.00000011
8	-0.00000005	0.00031905

Computed EigenVectors, Mass Normalized: 8

Node	Dx	Dy	Rz
Mode # 1	Freq (HZ) = 2.1417		
10	-0.000018	0.000000	-0.000000
20	-0.000018	0.000000	-0.000000
30	-0.000007	0.000000	-0.000000



40	0.000000	0.000000	-0.000000
50	0.001720	0.000000	-0.000022
60	0.004924	0.000000	-0.000030
70	0.008948	0.000000	-0.000036
80	0.013588	0.000000	-0.000041
90	0.018645	0.000000	-0.000043
100	0.023943	0.000000	-0.000045
110	0.028529	0.000000	-0.000045
120	0.031393	0.000000	-0.000045
130	0.032360	0.000000	-0.000045

Mode # 2 Freq (HZ) = 15.9519

10	-0.000096	-0.000000	-0.000000
20	-0.000096	-0.000000	-0.000000
30	-0.000036	-0.000000	-0.000000
40	0.000000	-0.000000	-0.000000
50	0.013102	-0.000000	-0.000070
60	0.023348	-0.000000	-0.000062
70	0.030262	-0.000000	-0.000023
80	0.030504	-0.000000	0.000038
90	0.021581	-0.000000	0.000104
100	0.004724	-0.000000	0.000152
110	-0.013203	-0.000000	0.000172
120	-0.024091	-0.000000	0.000172
130	-0.027768	-0.000000	0.000172

Mode # 3 Freq (HZ) = 25.8734

10	0.000000	0.000001	0.000000
20	0.000000	0.000001	0.000000
30	-0.000000	0.000001	0.000000
40	-0.000000	0.000001	-0.000000
50	-0.000000	0.011109	-0.000000
60	0.000000	0.014186	-0.000000
70	-0.000000	0.017109	0.000000
80	0.000000	0.019847	-0.000000
90	0.000000	0.022105	0.000000
100	-0.000000	0.023827	0.000000
110	0.000000	0.024805	0.000000
120	-0.000000	0.024823	0.000000
130	-0.000000	0.024830	0.000000

Mode # 4 Freq (HZ) = 44.6389

10	0.064284	-0.000000	0.000289
20	0.064264	-0.000000	0.000289
30	0.022840	-0.000000	0.000196
40	0.000002	-0.000000	0.000004
50	-0.009966	-0.000000	0.000014
60	-0.011594	-0.000000	-0.000011
70	-0.007760	-0.000000	-0.000040
80	-0.000322	-0.000000	-0.000046
90	0.005923	-0.000000	-0.000017
100	0.005170	-0.000000	0.000029
110	-0.000928	-0.000000	0.000055
120	-0.004460	-0.000000	0.000055
130	-0.005657	-0.000000	0.000055

Mode # 5 Freq (HZ) = 44.8606

10	0.017544	0.000000	0.000079
20	0.017539	0.000000	0.000079
30	0.006228	0.000000	0.000053
40	0.000001	0.000000	0.000000
50	0.036358	0.000000	-0.000051
60	0.042502	0.000000	0.000038
70	0.028692	0.000000	0.000146
80	0.001490	0.000000	0.000170
90	-0.021599	-0.000000	0.000065
100	-0.019004	-0.000000	-0.000104
110	0.003352	-0.000000	-0.000202
120	0.016299	-0.000000	-0.000203
130	0.020688	-0.000000	-0.000203



Mode # 6	Freq (HZ) = 73.7566		
10	0.000082	-0.000000	0.000000
20	0.000082	-0.000000	0.000000
30	0.000023	-0.000000	-0.000000
40	0.000001	-0.000000	-0.000001
50	0.044057	-0.000000	0.000058
60	0.029194	-0.000000	0.000181
70	-0.008374	-0.000000	0.000181
80	-0.029485	-0.000000	-0.000023
90	0.003470	-0.000000	-0.000143
100	0.028368	0.000000	0.000044
110	0.002166	0.000000	0.000246
120	-0.013862	0.000000	0.000248
130	-0.019327	0.000000	0.000248

Mode # 7	Freq (HZ) = 99.0817		
10	0.042810	-0.000000	0.000471
20	0.042774	-0.000000	0.000471
30	-0.036861	-0.000000	0.000165
40	-0.000005	-0.000000	-0.000001
50	-0.000005	-0.000008	-0.000000
60	-0.000008	-0.000010	-0.000000
70	-0.000007	-0.000010	-0.000000
80	0.000002	-0.000008	-0.000000
90	0.000004	-0.000003	0.000000
100	-0.000004	0.000003	0.000000
110	-0.000001	0.000007	-0.000000
120	0.000001	0.000007	-0.000000
130	0.000003	0.000007	-0.000000

Mode # 8	Freq (HZ) = 101.3930		
10	0.000003	0.000007	0.000000
20	0.000003	0.000007	0.000000
30	-0.000002	0.000005	0.000000
40	-0.000000	0.000003	-0.000000
50	-0.000000	0.026940	-0.000000
60	-0.000000	0.031731	-0.000000
70	-0.000000	0.031251	-0.000000
80	0.000000	0.025579	-0.000000
90	0.000000	0.010392	0.000000
100	-0.000000	-0.008660	-0.000000
110	-0.000000	-0.022143	-0.000000
120	0.000000	-0.022398	-0.000000
130	-0.000000	-0.022483	-0.000000

Mass Percentages:			
Included(X)	Included(Y)	Added(X)	Added(Y)
86.20	64.86	13.80	35.14

NOTE: In the following reports, "Max Contrib" displays the contribution of the Mode/Load combination having the maximum impact on the total (and names that Mode/Load combination).

Restraint Loads:			
Node	Fx(kgf)	Fy(kgf)	Mz(kgf-m.)
40	47850.0	26585.8	829294.4
Max	46225.1	24890.3	828600.4
Contrib	1 (X)	3 (Y)	1 (X)

Element Forces and Moments:			
Node	Fx(kgf)	Fy(kgf)	Mz(kgf-m.)
10	647.7	225.4	0.0
Max	523.4	225.4	0.0
Contrib	4 (X)	MM (Y)	



20	647.7	225.4	1.0
Max	523.4	225.4	0.8
Contrib	4 (X)	MM (Y)	4 (X)
20	3766.9	1311.1	1.0
Max	3044.4	1311.1	0.8
Contrib	4 (X)	MM (Y)	4 (X)
30	3766.9	1311.1	11301.8
Max	3044.4	1311.1	9134.1
Contrib	4 (X)	MM (Y)	4 (X)
30	8050.1	5594.4	11301.8
Max	6580.9	5594.4	9134.1
Contrib	4 (X)	MM (Y)	4 (X)
40	8050.1	5594.4	25344.7
Max	6580.9	5594.4	21308.8
Contrib	4 (X)	MM (Y)	4 (X)
40	46736.1	24893.5	828953.9
Max	46232.2	24889.5	828577.9
Contrib	1 (X)	3 (Y)	1 (X)
50	46736.1	24893.5	689905.3
Max	46232.2	24889.5	689881.2
Contrib	1 (X)	3 (Y)	1 (X)
50	46443.7	24285.9	689905.3
Max	46043.5	24284.2	689881.2
Contrib	1 (X)	3 (Y)	1 (X)
60	46443.7	24285.9	551929.5
Max	46043.5	24284.2	551750.6
Contrib	1 (X)	3 (Y)	1 (X)
60	45432.4	23074.7	551929.5
Max	45197.1	23073.0	551750.6
Contrib	1 (X)	3 (Y)	1 (X)
70	45432.4	23074.7	417012.6
Max	45197.1	23073.0	416159.3
Contrib	1 (X)	3 (Y)	1 (X)
70	43789.6	21616.3	417012.6
Max	43658.9	21612.2	416159.3
Contrib	1 (X)	3 (Y)	1 (X)
80	43789.6	21616.3	287281.3
Max	43658.9	21612.2	285182.4
Contrib	1 (X)	3 (Y)	1 (X)
80	38510.9	17837.5	287281.3
Max	38429.2	17817.9	285182.4
Contrib	1 (X)	3 (Y)	1 (X)
90	38510.9	17837.5	172816.5
Max	38429.2	17817.9	169894.6
Contrib	1 (X)	3 (Y)	1 (X)
90	31539.7	13633.1	172816.5
Max	31253.4	13592.1	169894.6
Contrib	1 (X)	3 (Y)	1 (X)
100	31539.7	13633.1	78616.3
Max	31253.4	13592.1	76134.4
Contrib	1 (X)	3 (Y)	1 (X)
100	22737.9	9124.9	78616.3
Max	22127.6	9081.1	76134.4
Contrib	1 (X)	3 (Y)	1 (X)
110	22737.9	9124.9	20685.5
Max	22127.6	9081.1	19709.0
Contrib	1 (X)	3 (Y)	1 (X)
110	11491.7	4278.2	20685.5
Max	10958.7	4257.4	19709.0
Contrib	1 (X)	3 (Y)	1 (X)



120	11491.7	4278.2	2299.4
Max	10958.7	4257.4	2175.1
Contrib	1 (X)	3 (Y)	1 (X)
120	4258.1	1542.7	2299.4
Max	4027.9	1535.1	2175.1
Contrib	1 (X)	3 (Y)	1 (X)
130	4258.1	1542.7	0.0
Max	4027.9	1535.1	0.0
Contrib	1 (X)	3 (Y)	

Displacements:

Node	Dx(mm)	Dy(mm)	Rz(Deg)
10	0.0173	0.0007	0.0002
Max	0.0136	0.0007	0.0001
Contrib	4 (X)	MM (Y)	4 (X)
20	0.0173	0.0007	0.0002
Max	0.0136	0.0007	0.0001
Contrib	4 (X)	MM (Y)	4 (X)
30	0.0062	0.0005	0.0001
Max	0.0048	0.0005	0.0001
Contrib	4 (X)	MM (Y)	4 (X)
40	0.0000	0.0000	0.0001
Max	0.0000	0.0000	0.0001
Contrib	1 (X)	3 (Y)	1 (X)
50	0.5901	0.0130	0.0172
Max	0.5894	0.0130	0.0172
Contrib	1 (X)	3 (Y)	1 (X)
60	1.6883	0.0165	0.0232
Max	1.6876	0.0165	0.0232
Contrib	1 (X)	3 (Y)	1 (X)
70	3.0672	0.0200	0.0279
Max	3.0665	0.0200	0.0279
Contrib	1 (X)	3 (Y)	1 (X)
80	4.6571	0.0231	0.0313
Max	4.6567	0.0231	0.0313
Contrib	1 (X)	3 (Y)	1 (X)
90	6.3897	0.0258	0.0336
Max	6.3895	0.0258	0.0335
Contrib	1 (X)	3 (Y)	1 (X)
100	8.2053	0.0278	0.0348
Max	8.2053	0.0278	0.0347
Contrib	1 (X)	3 (Y)	1 (X)
110	9.7767	0.0289	0.0352
Max	9.7766	0.0289	0.0351
Contrib	1 (X)	3 (Y)	1 (X)
120	10.7584	0.0290	0.0352
Max	10.7583	0.0290	0.0351
Contrib	1 (X)	3 (Y)	1 (X)
130	11.0898	0.0290	0.0352
Max	11.0896	0.0290	0.0351
Contrib	1 (X)	3 (Y)	1 (X)

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	268.174	...	647.681	...	0.33820E-08
20	30	35	656.643	3766.93	546.156	1.0282
30	40	207.5	2524.83	8050.12	5318.37	11301.8
40	50	150	19907.4	46736.1	186599	828954
50	60	450	14060.4	46443.7	141155	689905
60	70	750	12046.8	45432.4	101994	551930
70	80	1050	9987.23	43789.6	68942.8	417013
80	90	1350	7870.96	38510.9	42155.5	287281
90	100	1650	5695.66	31539.7	21805.5	172816
100	110	1927.5	3460.52	22737.9	8071.27	78616.3
110	120	2135	1524.92	11491.7	1709.99	20685.5
120	130	2242	457.995	4258.09	123.659	2299.38

Notes regarding vessels supported by Intermediate Skirts:

This model contains an intermediate skirt. The moment shown in the table above for the skirt, is the moment at the base of the skirt. It is computed by summing the moments framing into the top of the skirt and then by multiplying the total shear times the skirt length. The force on the skirt, times its moment arm is added to the result.

The moment at the top of the skirt is also computed. The stress at the skirt top is also computed and the worst of the stresses is displayed in the stress report.

The cumulative shear force on the skirt is generally the sum of the shear forces in the wind/seismic report including the shear force on the skirt and possibly other applied forces.

Bending Moments at top of Intermediate Skirt (Wind condition):

= [-Moment at Skirt top (below) + Moment at Skirt top (above)]
 = [-11106.4 + 141154.7]
 = 130048.4 kgf-m.

Moment at Skirt Top, operating case : 130048.4 kgf-m.

Bending Moments at top of Intermediate Skirt (Earthquake condition):

= [-Moment at Skirt top (below) + Moment at Skirt top (above)]
 = [-0.0 + 0.0]
 = 0.0 kgf-m.

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	5436.87	5199.46	342439808	328227968
20	30	10914.6	10678.7	685605760	672238144
30	40	10914.6	10678.7	685605760	672238144
40	50	2640.99	2640.99	180635024	180635024
50	60	10914.6	10678.7	685605760	672238144
60	70	10914.6	10678.7	685605760	672238144
70	80	10914.6	10678.7	685605760	672238144
80	90	10914.6	10678.7	685605760	672238144
90	100	10914.6	10678.7	685605760	672238144
100	110	10914.6	10678.7	685605760	672238144
110	120	10914.6	10678.7	685605760	672238144
120	130	10914.6	10678.7	685605760	672238144

Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1672.27	3002.91	-908.354	-1233.95
20	30	1672.27	3002.91	-999.196	-1273.61
30	40	1672.27	3002.91	-999.196	-1273.61
40	50	1386.78	2518.57	-794.046	-1120.19
50	60	1672.27	3002.91	-999.196	-1273.61
60	70	1672.27	3002.91	-999.196	-1273.61
70	80	1672.27	3002.91	-999.196	-1273.61
80	90	1672.27	3002.91	-999.196	-1273.61
90	100	1672.27	3002.91	-999.196	-1273.61
100	110	1672.27	3002.91	-999.196	-1273.61
110	120	1488.73	2663.87	-999.196	-1251.46
120	130	1488.73	2663.87	-999.196	-1251.46

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	1567.55	-11.1866	2385.55
20	30	753.585	-5.92542	1173.24
30	40	753.585	-5.92542	1173.24
40	50	...	...	...
50	60	753.585	-5.92542	1173.24
60	70	753.585	-5.92542	1173.24
70	80	753.585	-5.92542	1173.24
80	90	753.585	-5.92542	1173.24
90	100	753.585	-5.92542	1173.24
100	110	753.585	-5.92542	1173.24
110	120	...	...	...
120	130	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	...	...	...	...	...
20	30	0.88735	0.89934	...	-0.96318	-0.96318
30	40	4.27844	4.3328	...	-0.96318	-0.96318
40	50	-137.576	-138.671	...	0.43347	0.43347
50	60	-34.0243	-34.0243	...	0.11648	0.11648
60	70	-31.5228	-31.5228	...	0.11648	0.11648
70	80	-29.0213	-29.0213	...	0.11648	0.11648
80	90	-26.5199	-26.5199	...	0.11648	0.11648
90	100	-17.7147	-17.7147	...	0.11648	0.11648
100	110	-15.2132	-15.2132	...	0.11648	0.11648
110	120	-6.51205	-6.51205	...	...	...
120	130	-3.68415	-3.68415	...	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	...	...	0.10304E-11	...	...
20	30	...	0.081244	0.00015295	...	...
30	40	...	0.79114	1.68122	...	...
40	50	...	71.9951	...	...	...
50	60	...	20.9977	102.628	...	...
60	70	...	15.1723	82.1033	...	...
70	80	...	10.2557	62.0335	...	...
80	90	...	6.27092	42.7351	...	...
90	100	...	3.24372	25.7076	...	...
100	110	...	1.20066	11.6947	...	...
110	120	...	0.25437	3.07711	...	...
120	130	...	0.018395	0.34205	...	...

Longitudinal Stresses due to these Conditions:



From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	0.44402E-12
20	30	...	...	...	0.00014776
30	40	...	...	...	1.6356
40	50	...	...	...	...
50	60	...	...	...	101.917
60	70	...	...	...	81.5659
70	80	...	...	...	61.655
80	90	...	...	...	42.4916
90	100	...	...	...	25.5726
100	110	...	...	...	11.6389
110	120	...	...	...	3.06319
120	130	...	...	...	0.34082

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	...	-0.043344
20	30	...	-0.12278
30	40	...	-0.52388
40	50	...	-9.42582
50	60	...	-2.27423
60	70	...	-2.16081
70	80	...	-2.02424
80	90	...	-1.67037
90	100	...	-1.27666
100	110	...	-0.8545
110	120	...	-0.40063
120	130	...	-0.14446

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...
110	120	...	...	...	...
120	130	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...



60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	1672.27	0.00	908.35	0.0000	-0.0000	...
20	1.77	1672.27		1983.27	0.0011		...
30	4.45	1672.27		1983.27	0.0027		0.0026
40	210.00	1386.78	-210.00	1334.74	0.1514	0.1573	0.1158
50	55.14	1672.27	-55.14	1983.27	0.0330	0.0278	0.0203
60	46.81	1672.27	-46.81	1983.27	0.0280	0.0236	0.0188
70	39.39	1672.27	-39.39	1983.27	0.0236	0.0199	0.0173
80	32.91	1672.27	-32.91	1983.27	0.0197	0.0166	0.0158
90	21.07	1672.27	-21.07	1983.27	0.0126	0.0106	0.0106
100	16.53	1672.27	-16.53	1983.27	0.0099	0.0083	0.0091

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.04	1672.27	-0.04	1060.28	0.0000	0.0000	0.0155
20	1.73	1672.27		1983.27	0.0010		...



30	4.43	1672.27		1983.27	0.0026		0.0022
40	147.44	1386.78	-147.44	1334.74	0.1063	0.1105	0.1238
50	138.33	1672.27	-138.33	1983.27	0.0827	0.0697	0.0514
60	115.37	1672.27	-115.37	1983.27	0.0690	0.0582	0.0412
70	92.82	1672.27	-92.82	1983.27	0.0555	0.0468	0.0311
80	70.80	1672.27	-70.80	1983.27	0.0423	0.0357	0.0342
90	44.68	1672.27	-44.68	1983.27	0.0267	0.0225	0.0388
100	27.82	1672.27	-27.82	1983.27	0.0166	0.0140	0.0421

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	1672.27	0.00	1060.28	0.0000	-0.0000	0.0261
20	1.78	1672.27		1983.27	0.0011		...
30	4.50	1672.27		1983.27	0.0027		0.0026
40	211.10	1386.78	-211.10	1334.74	0.1522	0.1582	0.1168
50	55.14	1672.27	-55.14	1983.27	0.0330	0.0278	0.0203
60	46.81	1672.27	-46.81	1983.27	0.0280	0.0236	0.0188
70	39.39	1672.27	-39.39	1983.27	0.0236	0.0199	0.0173
80	32.91	1672.27	-32.91	1983.27	0.0197	0.0166	0.0158
90	21.07	1672.27	-21.07	1983.27	0.0126	0.0106	0.0106
100	16.53	1672.27	-16.53	1983.27	0.0099	0.0083	0.0091

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.04	1672.27	-0.04	1060.28	0.0000	0.0000	0.0155
20	1.74	1672.27		1983.27	0.0010		...
30	4.53	1672.27		1983.27	0.0027		0.0023
40	148.53	1386.78	-148.53	1334.74	0.1071	0.1113	0.1247
50	139.04	1672.27	-139.04	1983.27	0.0831	0.0701	0.0518
60	115.90	1672.27	-115.90	1983.27	0.0693	0.0584	0.0415
70	93.20	1672.27	-93.20	1983.27	0.0557	0.0470	0.0313
80	71.04	1672.27	-71.04	1983.27	0.0425	0.0358	0.0342
90	44.82	1672.27	-44.82	1983.27	0.0268	0.0226	0.0388
100	27.88	1672.27	-27.88	1983.27	0.0167	0.0141	0.0421

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	3002.91	0.00	1233.95	0.0000	-0.0000	0.0257
20	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
30	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
40	0.00	2518.57	0.00	1120.19	0.0000	-0.0000	0.0072
50	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	0.0011
60	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	0.0014
70	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	0.0017
80	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	0.0020
90	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	0.0023
100	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	0.0026

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	3002.91	0.00	1233.95	0.0000	-0.0000	...
20	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
30	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
40	0.00	2518.57	0.00	1120.19	0.0000	-0.0000	...
50	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
60	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
70	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
80	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
90	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...
100	0.00	3002.91	0.00	1273.61	0.0000	-0.0000	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW



From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1567.93	1672.27		908.35	0.9376		...
20	1373.63	1672.27		1983.27	0.8214		...
30	1373.44	1672.27		1983.27	0.8213		0.0026
40	211.10	1386.78	-211.10	1334.74	0.1522	0.1582	0.1168
50	1376.18	1672.27		1983.27	0.8229		0.0203
60	1375.55	1672.27		1983.27	0.8226		0.0188
70	1375.03	1672.27		1983.27	0.8223		0.0173
80	1374.59	1672.27		1983.27	0.8220		0.0158
90	1373.95	1672.27		1983.27	0.8216		0.0106
100	1373.68	1672.27		1983.27	0.8214		0.0091

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1567.91	1672.27		908.35	0.9376		...
20	1373.64	1672.27		1983.27	0.8214		...
30	1373.47	1672.27		1983.27	0.8213		0.0023
40	148.53	1386.78	-148.53	1334.74	0.1071	0.1113	0.1247
50	1384.45	1672.27		1983.27	0.8279		0.0518
60	1381.59	1672.27		1983.27	0.8262		0.0415
70	1379.14	1672.27		1983.27	0.8247		0.0313
80	1377.11	1672.27		1983.27	0.8235		0.0342
90	1375.19	1672.27		1983.27	0.8223		0.0388
100	1374.18	1672.27		1983.27	0.8217		0.0421

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	11.19	1672.27	-11.19	1060.28	0.0067	0.0106	0.0103
20	5.91	1672.27	-4.14	1983.27	0.0035	0.0021	0.0140
30	1.76	1672.27	-1.42	1983.27	0.0011	0.0007	0.0162
40	211.10	1386.78	-211.10	1334.74	0.1522	0.1582	0.1168
50	61.06	1672.27	-61.06	1983.27	0.0365	0.0308	0.0259
60	52.74	1672.27	-52.74	1983.27	0.0315	0.0266	0.0236
70	45.32	1672.27	-45.32	1983.27	0.0271	0.0229	0.0221
80	38.83	1672.27	-38.83	1983.27	0.0232	0.0196	0.0209
90	27.00	1672.27	-27.00	1983.27	0.0161	0.0136	0.0189
100	22.46	1672.27	-22.46	1983.27	0.0134	0.0113	0.0183

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	11.23	1672.27	-11.23	1060.28	0.0067	0.0106	0.0099
20	6.11	1672.27	-4.19	1983.27	0.0037	0.0021	0.0139
30	2.83	1672.27	-2.83	1983.27	0.0017	0.0014	0.0160
40	148.53	1386.78	-148.53	1334.74	0.1071	0.1113	0.1247
50	144.97	1672.27	-144.97	1983.27	0.0867	0.0731	0.0657
60	121.83	1672.27	-121.83	1983.27	0.0729	0.0614	0.0546
70	99.12	1672.27	-99.12	1983.27	0.0593	0.0500	0.0438
80	76.97	1672.27	-76.97	1983.27	0.0460	0.0388	0.0342
90	50.74	1672.27	-50.74	1983.27	0.0303	0.0256	0.0388
100	33.80	1672.27	-33.80	1983.27	0.0202	0.0170	0.0421

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	2453.46	3002.91		1233.95	0.8170		0.0145
20	2256.22	3002.91		1273.61	0.7513		...
30	2256.22	3002.91		1273.61	0.7513		...
40	0.00	2518.57	0.00	1120.19	0.0000	-0.0000	0.0072
50	2256.22	3002.91		1273.61	0.7513		0.0011
60	2256.22	3002.91		1273.61	0.7513		0.0014
70	2256.22	3002.91		1273.61	0.7513		0.0017



80	2256.22	3002.91	1273.61	0.7513	0.0020
90	2256.22	3002.91	1273.61	0.7513	0.0023
100	2256.22	3002.91	1273.61	0.7513	0.0026

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	2453.46	3002.91		1233.95	0.8170		...
20	2256.22	3002.91		1273.61	0.7513		...
30	2256.22	3002.91		1273.61	0.7513		...
40	0.00	2518.57	0.00	1120.19	0.0000	-0.0000	...
50	2256.22	3002.91		1273.61	0.7513		...
60	2256.22	3002.91		1273.61	0.7513		...
70	2256.22	3002.91		1273.61	0.7513		...
80	2256.22	3002.91		1273.61	0.7513		...
90	2256.22	3002.91		1273.61	0.7513		...
100	2256.22	3002.91		1273.61	0.7513		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1567.93	1672.27		908.35	0.9376		...
20	1373.63	1672.27		1983.27	0.8214		...
30	1373.46	1672.27		1983.27	0.8213		0.0026
40	138.01	1386.78	-138.01	1334.74	0.0995	0.1034	0.1158
50	1374.90	1672.27		1983.27	0.8222		0.0203
60	1374.69	1672.27		1983.27	0.8220		0.0188
70	1374.48	1672.27		1983.27	0.8219		0.0173
80	1374.28	1672.27		1983.27	0.8218		0.0158
90	1373.81	1672.27		1983.27	0.8215		0.0106
100	1373.63	1672.27		1983.27	0.8214		0.0091

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1567.93	1672.27		908.35	0.9376		...
20	1373.61	1672.27		1983.27	0.8214		...
30	1373.44	1672.27		1983.27	0.8213		0.0025
40	138.67	1386.78	-138.67	1334.74	0.1000	0.1039	0.1168
50	1374.85	1672.27		1983.27	0.8221		0.0199
60	1374.65	1672.27		1983.27	0.8220		0.0184
70	1374.44	1672.27		1983.27	0.8219		0.0170
80	1374.25	1672.27		1983.27	0.8218		0.0155
90	1373.79	1672.27		1983.27	0.8215		0.0104
100	1373.61	1672.27		1983.27	0.8214		0.0089

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1567.93	1672.27		908.35	0.9376		...
20	1373.63	1672.27		1983.27	0.8214		...
30	1373.46	1672.27		1983.27	0.8213		0.0026
40	139.10	1386.78	-139.10	1334.74	0.1003	0.1042	0.1168
50	1374.90	1672.27		1983.27	0.8222		0.0203
60	1374.69	1672.27		1983.27	0.8220		0.0188
70	1374.48	1672.27		1983.27	0.8219		0.0173
80	1374.28	1672.27		1983.27	0.8218		0.0158
90	1373.81	1672.27		1983.27	0.8215		0.0106
100	1373.63	1672.27		1983.27	0.8214		0.0091

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1567.93	1672.27		908.35	0.9376		...
20	1373.63	1672.27		1983.27	0.8214		...



30	1373.46	1672.27		1983.27	0.8213		0.0026
40	138.01	1386.78	-138.01	1334.74	0.0995	0.1034	0.1158
50	1374.90	1672.27		1983.27	0.8222		0.0203
60	1374.69	1672.27		1983.27	0.8220		0.0188
70	1374.48	1672.27		1983.27	0.8219		0.0173
80	1374.28	1672.27		1983.27	0.8218		0.0158
90	1373.81	1672.27		1983.27	0.8215		0.0106
100	1373.63	1672.27		1983.27	0.8214		0.0091

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	1672.27	0.00	908.35	0.0000	-0.0000	...
20	1.86	1672.27		1983.27	0.0011		...
30	5.30	1672.27		1983.27	0.0032		0.0026
40	139.10	1386.78	-139.10	1334.74	0.1003	0.1042	0.1168
50	34.14	1672.27	-34.14	1983.27	0.0204	0.0172	0.0203
60	31.64	1672.27	-31.64	1983.27	0.0189	0.0160	0.0188
70	29.14	1672.27	-29.14	1983.27	0.0174	0.0147	0.0173
80	26.64	1672.27	-26.64	1983.27	0.0159	0.0134	0.0158
90	17.83	1672.27	-17.83	1983.27	0.0107	0.0090	0.0106
100	15.33	1672.27	-15.33	1983.27	0.0092	0.0077	0.0091

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1567.91	1672.27		1060.28	0.9376		0.0143
20	1373.64	1672.27		1983.27	0.8214		...
30	1373.47	1672.27		1983.27	0.8213		0.0023
40	148.53	1386.78	-148.53	1334.74	0.1071	0.1113	0.1247
50	1375.02	1672.27		1983.27	0.8222		0.0217
60	1374.80	1672.27		1983.27	0.8221		0.0201
70	1374.58	1672.27		1983.27	0.8220		0.0185
80	1374.36	1672.27		1983.27	0.8219		0.0168
90	1373.86	1672.27		1983.27	0.8216		0.0113
100	1373.66	1672.27		1983.27	0.8214		0.0096

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	11.23	1672.27	-11.23	1060.28	0.0067	0.0106	0.0103
20	6.11	1672.27	-4.19	1983.27	0.0037	0.0021	...
30	3.08	1672.27	-1.15	1983.27	0.0018	0.0006	0.0030
40	148.53	1386.78	-148.53	1334.74	0.1071	0.1113	0.1247
50	42.34	1672.27	-42.34	1983.27	0.0253	0.0213	0.0217
60	39.73	1672.27	-39.73	1983.27	0.0238	0.0200	0.0201
70	37.09	1672.27	-37.09	1983.27	0.0222	0.0187	0.0185
80	34.23	1672.27	-34.23	1983.27	0.0205	0.0173	0.0168
90	25.03	1672.27	-25.03	1983.27	0.0150	0.0126	0.0113
100	22.11	1672.27	-22.11	1983.27	0.0132	0.0111	0.0096

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1567.93	1672.27		1060.28	0.9376		0.0096
20	1373.63	1672.27		1983.27	0.8214		...
30	1373.46	1672.27		1983.27	0.8213		0.0026
40	139.10	1386.78	-139.10	1334.74	0.1003	0.1042	0.1168
50	1374.90	1672.27		1983.27	0.8222		0.0203
60	1374.69	1672.27		1983.27	0.8220		0.0188
70	1374.48	1672.27		1983.27	0.8219		0.0173
80	1374.28	1672.27		1983.27	0.8218		0.0158
90	1373.81	1672.27		1983.27	0.8215		0.0106
100	1373.63	1672.27		1983.27	0.8214		0.0091

Absolute Maximum of the all of the Stress Ratio's

0.9376



Governing Element: Bottom head
 Governing Load Case 7 : IP+OW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom head	1567.6	...	...	...	1567.6	+ Tens
Bottom head	1567.6	...	...	...	...	- Comp
Shell-1	753.6	0.9	-1.0	0.1	753.6	+ Tens
Shell-1	753.6	0.9	1.0	-0.1	...	- Comp
Shell-2	753.6	4.3	-1.0	0.8	757.7	+ Tens
Shell-2	753.6	4.3	1.0	-0.8	...	- Comp
Inte skirt	...	-138.7	0.4	72.0	...	+ Tens
Inte skirt	...	-138.7	-0.4	-72.0	-211.1	- Comp
Shell-3	753.6	-34.0	0.1	21.0	740.7	+ Tens
Shell-3	753.6	-34.0	-0.1	-21.0	...	- Comp
Shell-4	753.6	-31.5	0.1	15.2	737.4	+ Tens
Shell-4	753.6	-31.5	-0.1	-15.2	...	- Comp
Shell-5	753.6	-29.0	0.1	10.3	734.9	+ Tens
Shell-5	753.6	-29.0	-0.1	-10.3	...	- Comp
Shell-6	753.6	-26.5	0.1	6.3	733.5	+ Tens
Shell-6	753.6	-26.5	-0.1	-6.3	...	- Comp
Shell-7	753.6	-17.7	0.1	3.2	739.2	+ Tens
Shell-7	753.6	-17.7	-0.1	-3.2	...	- Comp
Shell-8	753.6	-15.2	0.1	1.2	739.7	+ Tens
Shell-8	753.6	-15.2	-0.1	-1.2	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length cm
- Fxa - Allowable comp mem stress due to compression with lamc ≤ 0.15 , kgf/cm²
- Fba - Allowable all comp mem stress due to bending moment, kgf/cm²
- Fva - Allowable shear stress when only shear is present, kgf/cm²
- Fha - Allowable hoop comp memb stress under ext pressure alone, kgf/cm²
- Fxha - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , kgf/cm²
- Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , kgf/cm²
- Faha - Allowable axial comp w/hoop compression and lamc > 0.15 , kgf/cm²
- Fbha - Allowable axial comp from bending w/hoop compression, kgf/cm²
- Fhba - Allowable hoop comp w/long compression due to moment, kgf/cm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
------	------	---	-----	-----	-----	-----	------	------	------	------	------



10						1060	70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1060	70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1060	70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1079	70307		70307	70307	70307
20	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
30	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
40	0.566	22819	1367	1373	618		70307	1367	70307	70307	70307
50	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
60	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
70	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
80	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
90	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
100	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
110											
120											

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
30	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
40	0.566	22819	1367	1373	618		70307	1367	70307	70307	70307
50	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
60	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
70	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
80	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
90	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
100	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307



110
120

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						70307		70307		70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307

110
120

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						70307		70307		70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307

110
120

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1060	70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	492	965	415	80	873
30	0.733	22819	1983	1983	1278	878	813	998	685	135	869
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	2241	651	1888	1265	676
60	0.733	22819	1983	1983	1278	878	2218	688	1868	1013	749
70	0.733	22819	1983	1983	1278	878	2185	728	1840	736	802
80	0.733	22819	1983	1983	1278	878	2140	770	1803	474	838
90	0.733	22819	1983	1983	1278	878	1840	920	1550	255	860
100	0.733	22819	1983	1983	1278	878	1700	956	1432	101	871

110
120

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1060	70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	481	963	405	74	873
30	0.733	22819	1983	1983	1278	878	764	994	643	202	864
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	2256	619	1901	1849	203
60	0.733	22819	1983	1983	1278	878	2238	656	1886	1818	249
70	0.733	22819	1983	1983	1278	878	2212	696	1863	1768	321
80	0.733	22819	1983	1983	1278	878	2172	742	1829	1667	439
90	0.733	22819	1983	1983	1278	878	1901	900	1601	1410	617
100	0.733	22819	1983	1983	1278	878	1751	945	1475	823	787

110
120

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1504	70307		70307	70307	70307
20	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
30	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
40	0.566	22819	1367	1373	618		70307	1367	70307	70307	70307
50	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307



60	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
70	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
80	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
90	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
100	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
110											
120											

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
30	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
40	0.566	22819	1367	1373	618		70307	1367	70307	70307	70307
50	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
60	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
70	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
80	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
90	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
100	0.706	22819	1995	1995	1290	901	70307	1995	70307	70307	70307
110											
120											

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
------	------	---	-----	-----	-----	-----	------	------	------	------	------



10							70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1060	70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
110											
120											

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1060	70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	481	963	405	70307	70307
30	0.733	22819	1983	1983	1278	878	764	994	643	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	2256	619	1901	70307	70307
60	0.733	22819	1983	1983	1278	878	2238	656	1886	70307	70307
70	0.733	22819	1983	1983	1278	878	2212	696	1863	70307	70307
80	0.733	22819	1983	1983	1278	878	2172	742	1829	70307	70307
90	0.733	22819	1983	1983	1278	878	1901	900	1601	70307	70307
100	0.733	22819	1983	1983	1278	878	1751	945	1475	70307	70307
110											
120											

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1060	70307		70307	70307	70307
20	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
30	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
40	0.586	22819	1334	1340	608		70307	1334	70307	70307	70307
50	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
60	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
70	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
80	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
90	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307
100	0.733	22819	1983	1983	1278	878	70307	1983	70307	70307	70307



110
120

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Inte skirt

Mean skirt diameter at the critical height is computed later.
 This is a Conical type Support Skirt , Wider at the Bottom

Half Apex Angle of the skirt	a	9.267 deg
Total vessel (empty) weight	Fg	554891.938 kgf
Weight of vessel contents (liquid)	Ff	3750.374 kgf
Vessel Gross Weight	Fg + Ff	558642.312 kgf
Skirt Mean Diameter Top End	DzTop	2765.604 mm
Skirt Mean Diameter Bottom End	DzBot	3744.603 mm
Skirt Analysis Thickness (Corroded)	ez	30.000 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	0.000 kgf
Moment at section 1-1	M1	689905.312 kgf-m.
Applied Moment - Bottom of Skirt	Mb	828953.938 kgf-m.
Weight at Bottom of Skirt	W	567033.062 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:



For a Type B Skirt, the next element must be a dished head and this is not the case. Please refer to Figure 16.12-2 of EN 13445-3 for more information.

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Shop/Field Installation Options :

Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	1157.492 cm
Center of Gravity of the Insulation	1137.391 cm
Center of Gravity of the Nozzles	1231.573 cm
Center of Gravity of the Added Weights (Operating)	1133.726 cm
Center of Gravity of the Added Weights (Empty)	1133.726 cm
Center of Gravity of Bare Shell New and Cold	1384.955 cm
Center of Gravity of Bare Shell Corroded	1389.291 cm
Vessel CG in the Operating Condition	1258.697 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	1259.232 cm
Vessel CG in the Test Condition	1240.984 cm

Rigging Analysis Results:

Because this model is supported by an intermediate skirt, the rigging analysis cannot be performed.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3774.2063	mm
Skirt Thickness	STHK	30.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.0000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	1784.41	kgf/cm ²
Basing Yield Stress		2671.52	kgf/cm ²
Inside Diameter of Basing	DI, D4	3603.0002	mm
Outside Diameter of Basing	DOU, D5	4008.9998	mm
Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Root Area of a Single Bolt	Area	10.8300	cm ²
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	1400.02	kgf/cm ²
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	3895.0002	mm
Bolt Allowable Shear Stress		1099.960	kgf/cm ²
Ultimate Comp. Strength of Concrete	FPC	102.0	kgf/cm ²
Allowable Comp. Strength of Concrete	FC	89.2	kgf/cm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	102.0000	mm
Width of Gussets at Base Plate	BWDT	102.0000	mm
Gusset Plate Elastic Modulus	E	2057170.1	kgf/cm ²
Gusset Plate Yield Stress	SY	2671.5	kgf/cm ²
Height of Gussets	HG, h1	204.0000	mm
Distance between Gussets	RG, b6	102.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	38.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	102.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	66.0000	mm
External Corrosion Allowance	CA	0.0000	mm
Dead Weight of Vessel	DW	563282.7	kgf
Operating Weight of Vessel	ROW	567033.1	kgf
Earthquake Moment on Basing	EQMOM	836211.9	kgf-m.
Wind Moment on Basing	WIMOM	193856.7	kgf-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 836212/3895.0) - 567033) / 16$$

$$= 18232.5 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 193857/3895.0) - 563283) / 16$$

$$= -22762.5 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3895.0) - 567033) / 16$$

$$= -35439.6 \text{ kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	10.8300	cm ²
Area Available in all the Bolts	Abss * RN :	173.2800	cm ²



*There seems to be no tensile load on bolts per the Brownell and Young Method.
 It is possible that this method is encountering a problem. It is strongly
 recommended that you try the simplified analysis method for the basing.*

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 43.005[(2 \cdot 1.0 \cdot 3895.0 + 203.0) / (2 \cdot 1.0 \cdot 3895.0)]$$

$$= 44.126 \text{ kgf/cm}^2, \text{ Should be } \leq 89.2 \text{ kgf/cm}^2$$

Values for table 10.3, l = 117.397, b = 382.391, l/b = 0.307007

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(463.525, 2637.404) = 2637.404 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow) + Ca}$$

$$= \sqrt{(6 \cdot 2637.404 / 1763.2) + 0.0}$$

$$= 29.958 \text{ mm}$$

Nomenclature:

a = (Dc-Ds)/2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 102.0, b = 102.0, b/l = 1.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi a/l)) / (\pi g))) + 1] - g1 \cdot P / (4\pi)$$

$$= 15162.22 / (4 \cdot \pi) [1.3(\ln((2 \cdot 102.0 \cdot \sin(\pi \cdot 60.397/102.0)) / (\pi \cdot 34.925))) + 1]$$

$$- 0.583 \cdot 15162.217 / (4 \cdot \pi)]$$

$$= 1408.9056 \text{ kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb) + Ca}$$

$$= \sqrt{(6 \cdot Abs(1408.91) / 1781.01) + 0.0}$$

$$= 21.7863 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot lavg) + Ca$$

$$= 15162.22 / (1265.526 \cdot 102.0) + 0.0$$

$$= 11.746 \text{ (not less than } 9.525 + 0.0) \text{ mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3895.0 / 16 = 764.781 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:
 $= 1.76(P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot Sklope))^{2/3} \cdot r^{1/3} + Ca$
 $= 1.76(15162 \cdot 60.397 / (764.78 \cdot 242.0 \cdot 2080))^{2/3} \cdot 1887.11^{1/3} + Ca$
 $= 8.35 + 0.0 = 8.35 \text{ mm}$

Shear Stress in a Single Bolt [tb]:
 $= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 46736.1 / (10.83 \cdot 16)$
 $= 269.7 \text{ kgf/cm}^2$. Must be less than 1100.0 kgf/cm².

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	29.9580	mm
Actual Basing Thickness as entered by user	38.0000	mm
Required Thickness of Chair Cap	21.7863	mm
Actual Top Ring Thickness as entered by user	38.0000	mm
Required Gusset thickness, + CA	11.7461	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	8.3495	mm
Given Thickness of Skirt	30.0000	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
 $= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$
 $= 211.099 \cdot (30.0 - 0.0)$
 $= 63.33 \text{ kgf/mm}$

Results for Computed Minimum Basing Weld Size [BWeld]:
 $= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 63 / [(0.4 \cdot 2080) \cdot 2 \cdot 0.707]$
 $= 5.383 \text{ mm}$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
 $= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (Hg + Tta))$
 $= 15162.2 / (58.0 + 2 \cdot (204.0 + 38.0))$
 $= 27.975 \text{ kgf/mm}$

Horizontal Plate Load [Wh]:
 $= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$
 $= 15162.2 \cdot 60.397 / (58.0 \cdot (242.0) + 0.6667 \cdot (242.0)^2)$
 $= 17.253 \text{ kgf/mm}$

Resultant Weld Load [Wr]:
 $= \sqrt{ Wv^2 + Wh^2 }$
 $= \sqrt{ 27.97^2 + 17.25^2 }$
 $= 32.867 \text{ kgf/mm}$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
 $= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 32.87 / [(0.4 \cdot 2080) \cdot 2 \cdot 0.707]$
 $= 2.794 \text{ mm}$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
 $= \text{Bolt Load} / (2 \cdot \text{TopWth})$
 $= 15162.2 / (2 \cdot 102.0)$
 $= 74.325 \text{ kgf/mm}$

Weld Load [Wh]:
 $= \text{Bolt Load} \cdot e / (2 \cdot Hgt \cdot \text{TopWth})$
 $= 15162.2 \cdot 60.4 / (2 \cdot 242.0 \cdot 102.0)$
 $= 18.549 \text{ kgf/mm}$



Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{W_v^2 + W_h^2} \\ &= \sqrt{74.32^2 + 18.55^2} \\ &= 76.604 \text{ kgf/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 76.6 / [(0.4 \cdot 2080) \cdot 2 \cdot 0.707] \\ &= 6.511 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.5110	mm
Required Gusset to Top Plate Double Fillet Weld Size	6.5110	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: B 450mm

From: 10

Pressure for Reinforcement Calculations	P	163.871	kgf/cm ²
Temperature for Internal Pressure	Temp	300	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	300	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1672.27	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Hemispherical Head	D	2516.00	mm
Head Finished (Minimum) Thickness	t	67.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1723.24	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		450.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	36.0000	mm
Flange Material		SA-182 F22	
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	94.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		130.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	67.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.10	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B 450mm

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	450.000	mm.
Actual Thickness Used in Calculation	36.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot \left(\exp \left(\frac{P}{S \cdot E} \right) - 1 \right) + c_i + c_o$$

$$= 0.5 \cdot 456.0 \cdot \left(\exp \left(\frac{163.871}{(1723.24 \cdot 1.0)} \right) - 1 \right) + 3.0 + 0.0$$

$$= 22.746 + 3.0 + 0.0 = 25.746 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln \left(\frac{2 \cdot t + D}{D} \right) - P_{liq}$$

$$= 1723.24 \cdot 1.0 \cdot \ln \left(\frac{2 \cdot 33.0 + 456.0}{456.0} \right) - 0.0$$

$$= 232.938 - 0.0 = 232.9384 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 66.336 mm
 Min. required shell thickness, external p + c : 9.604 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 35.992 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 700.7,	Allowable	: 1723.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 4091.7 kgf/cm ²	Passed
Occasional	: 527.9,	Allowable	: 2291.9 kgf/cm ²	Passed
Shear	: 113.3,	Allowable	: 1206.3 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1723.2/1672.3, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 130.0 - 3.0$$

$$= 127.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 36.0 - 3.0$$

$$= 33.000 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2516.0/2 + 3.0$$

$$= 1261.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min \left(\sqrt{Reff \cdot t}, 2 \cdot R_n \right)$$

$$= \min \left(\sqrt{1261.0 \cdot 64.0}, 2 \cdot 228.0 \right)$$

$$= 284.085 \text{ mm}$$

Intermediate value [Cn]:

$$= \min \left(\left(\frac{t + t_e}{t_n} \right)^{0.35}, 1 \right)$$

$$= \min \left(\left(\frac{64.0 + 0.0}{127.0} \right)^{0.35}, 1 \right)$$

$$= \min(0.787, 1)$$

$$= 0.787$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:



$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(64.0 + 0.0 + 0.787 \cdot \sqrt{(228.0 \cdot 127.0)}, 800.0 + 64.0)$$

$$= 197.875 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 64.0 + ((0.0 \cdot 1.0) / 284.085)$$

$$= 64.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 64.0 \cdot 284.085$$

$$= 181.814 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 127.0 \cdot 197.875$$

$$= 251.302 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 181.814 + 1.0(251.302 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 433.616 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 127.0 / \ln[1 + 127.0/228.0]$$

$$= 286.829 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 64.0 / \ln[1 + 64.0/1261.0]$$

$$= 1292.736 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1672.27 \cdot 1.0$$

$$= 2508.4 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 163.871 \cdot 286.829 \cdot 197.875$$

$$= 93007.4 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 163.871 \cdot 1292.736 \cdot (284.085 + 127.0)/2$$

$$= 435424.8 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (163.871 \cdot 1292.736 \cdot 228.0)/2$$

$$= 241499.9 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (435424.8 + 241499.9 + 93007.4)/163.871$$

$$= 4698.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/(2 \cdot t_{eff}))$$

$$= 2508.405 / (2 \cdot 4698.404/433.616 - 1292.736/(2 \cdot 64.0))$$

$$= 216.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 1672.27[64.0/1292.736]$$

$$= 165.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(216.778, 165.579)$$

$$= 165.579 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (93007.398 + 435424.844 + 241499.922) / 433.616$$

$$= 1775.611 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 163.871 \cdot 1292.736 / (2 \cdot 64.0)$$

$$= 1655.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1775.611 - 1655.018, 1655.018)$$

$$= 1896.2 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2508.41	kgf/cm ²
Local Primary Membrane Stress	PL	1896.20	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	165.58	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (228.0 + 127.0) / 228.0$$

$$= 1.557 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (228.0 + 127.0)$$

$$= 557.633 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 127.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(241500 \cdot 1.56, 1.5 \cdot 1723.2(297.389 + 0.0), \pi/4 \cdot 163.9 \cdot 228.0^2 \cdot 1.56^2)$$

$$= 162199.234 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 162199.234 / (557.633(0.49 \cdot 7.071 + 0.6 \cdot 64.0 + 0.49 \cdot 0.0))$$

$$= 694.788 < \text{or} = \text{to } 1672.270 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 127.0 / \ln[1 + 127.0/228.0]$$

$$= 286.829 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 64.0 / \ln[1 + 64.0/1261.0]$$

$$= 1292.736 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1672.27$$

$$= 2508.4 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 286.829 \cdot 197.875$$

$$= 598.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n) / 2$$

$$= 1.055 \cdot 1292.736 \cdot (284.085 + 127.0) / 2$$

$$= 2802.1 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (1.055 \cdot 1292.736 \cdot 228.0) / 2$$

$$= 1554.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2802.1 + 1554.1 + 598.5) / 1.055$$

$$= 4698.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 2508.405 / (2 \cdot 4698.404 / 433.616 - 1292.736 / (2 \cdot 64.0))$$

$$= 216.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / R_{xs}]$$

$$= 2 \cdot 1672.27[64.0 / 1292.736]$$

$$= 165.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(216.778, 165.579)$$

$$= 165.579 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (598.533 + 2802.099 + 1554.13) / 433.616$$

$$= 11.427 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.055 \cdot 1292.736 / (2 \cdot 64.0)$$

$$= 10.7 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 11.427 - 10.651, 10.651)$$

$$= 12.2 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2508.41	kgf/cm ²
Local Primary Membrane Stress	PL	12.20	kgf/cm ²
Maximum Allowable External Pressure	Pmax	165.58	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-25 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	3 °C



Temperature reduction per Graph 3.13 : 28 °C
 The computed stress ratio is : 0.558

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (18.407 \cdot 1.0) / (36.0 - 3.0 - 0.0)$
 $= 0.558$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : 2 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 14 °C
 Temperature reduction per Graph 3.13 : 12 °C
 The computed stress ratio is : 0.780

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (49.914 \cdot 1.0) / (67.0 - 3.0 - 0.0)$
 $= 0.780$

Gov. MDMT of the nozzle to shell joint welded assembly : 2 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -39 °C

The Drop for this Nozzle is : 51.1285 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 918.1284 mm

Input Echo, WRC107/537 Item 1, Description: B 450mm :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	2516.000	mm
Vessel Thickness	Tv	67.000	mm
Design Temperature	T1	300.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	2110.70	kgf/cm²
Vessel Hot S.I. Allowable	Smh	1672.27	kgf/cm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	450.000	mm
Nozzle Thickness	Tn	130.000	mm
Nozzle Material		SA-336 F22	
Nozzle UNS Number		K21590	
Nozzle Cold S.I. Allowable	SNmc	2110.70	kgf/cm²
Nozzle Hot S.I. Allowable	SNmh	1723.24	kgf/cm²
Design Internal Pressure	Dp	163.871	kgf/cm²
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

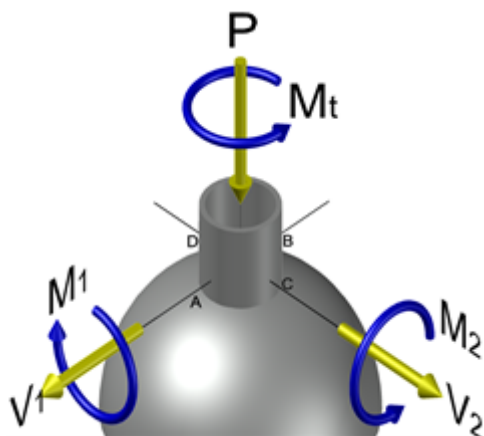
Radial Load (SUS)	P	-6373.6	kgf
Longitudinal Shear (SUS)	(V1) V1	5098.8	kgf
Circumferential Shear (SUS)	(Vc) V2	5098.8	kgf
Circumferential Moment (SUS)	(Mc) M1	7645.2	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	7645.2	kgf-m.

Torsional Moment (SUS) Mt 8919.4 kgf-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 710.0 + 2 \cdot 1.65 \cdot \sqrt{(1293.0 (67.0 - 3.0))}$$

$$= 1659.299 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-6373.6	kgf
Circumferential Shear	(VC) V2	5098.8	kgf
Longitudinal Shear	(VL) V1	5098.8	kgf
Circumferential Moment	(MC) M1	7645.2	kgf-m.
Longitudinal Moment	(ML) M2	7645.2	kgf-m.
Torsional Moment	MT	8919.4	kgf-m.

Dimensionless Param: U = 1.23 TAU = 5.00 (2.30) RHO = 0.50

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03280	(A,B,C,D)
$M(x) / P$	SP 1	0.03224	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03563	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.05061	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03563	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.05061	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05105	(A,B,C,D)
$M(y) / P$	SP 1	0.00926	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04985	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01627	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04985	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01627	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Rad. Bend. P		30.1	-30.1	30.1	-30.1	30.1	-30.1	30.1	-30.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-23.1	-23.1	23.1	23.1
Rad. Bend. MC		0.0	0.0	0.0	0.0	-197.0	197.0	197.0	-197.0
Rad. Memb. ML		-23.1	-23.1	23.1	23.1	0.0	0.0	0.0	0.0
Rad. Bend. ML		-197.0	197.0	197.0	-197.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-185.0	148.9	255.4	-198.9	-185.0	148.9	255.4	-198.9
Tang. Memb. P		7.9	7.9	7.9	7.9	7.9	7.9	7.9	7.9
Tang. Bend. P		8.6	-8.6	8.6	-8.6	8.6	-8.6	8.6	-8.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-32.3	-32.3	32.3	32.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-63.3	63.3	63.3	-63.3
Tang. Memb. ML		-32.3	-32.3	32.3	32.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-63.3	63.3	63.3	-63.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-79.1	30.3	112.3	-31.7	-79.1	30.3	112.3	-31.7
Shear VC		7.1	7.1	-7.1	-7.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-7.1	-7.1	7.1	7.1
Shear MT		17.6	17.6	17.6	17.6	17.6	17.6	17.6	17.6
Tot. Shear		24.7	24.7	10.5	10.5	10.5	10.5	24.7	24.7
Str. Int.		190.5	153.9	256.1	199.6	186.0	149.8	259.5	202.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1655.0	1655.0	1655.0	1655.0	1655.0	1655.0	1655.0	1655.0
Rad. Pl (SUS)		-18.0	-18.0	28.2	28.2	-18.0	-18.0	28.2	28.2
Rad. Q (SUS)		-166.9	166.9	227.1	-227.1	-166.9	166.9	227.1	-227.1
Long. Pm (SUS)		1655.0	1655.0	1655.0	1655.0	1655.0	1655.0	1655.0	1655.0
Long. Pl (SUS)		-24.4	-24.4	40.3	40.3	-24.4	-24.4	40.3	40.3
Long. Q (SUS)		-54.7	54.7	72.0	-72.0	-54.7	54.7	72.0	-72.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		7.1	7.1	-7.1	-7.1	-7.1	-7.1	7.1	7.1
Shear Q (SUS)		17.6	17.6	17.6	17.6	17.6	17.6	17.6	17.6
Pm (SUS)		1655.0	1655.0	1655.0	1655.0	1655.0	1655.0	1655.0	1655.0
Pm+Pl (SUS)		1641.6	1641.6	1698.6	1698.6	1641.6	1641.6	1698.6	1698.6
Pm+Pl+Q (Total)		1581.4	1808.9	1911.1	1624.0	1577.0	1804.9	1914.5	1626.9

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1655.02	1672.27	Passed
Pm+Pl (SUS)	1698.62	2508.41	Passed
Pm+Pl+Q (TOTAL)	1914.53	5017.04	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or



per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+PI+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: Top plug weight

From: 120

Pressure for Reinforcement Calculations	P	163.794	kgf/cm ²
Temperature for Internal Pressure	Temp	300	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	300	°C
Parent Material		SA-336 F11	+IT
Parent Allowable Stress at Temperature	Sv	1488.73	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1876.18	kgf/cm ²
Outside Diameter of Bolted Blind Flange	D	3070.00	mm
Head Finished (Minimum) Thickness	t	540.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	+IT
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1723.26	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.74	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	11.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	25.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	9.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: Top plug weight

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	11.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.0 \cdot (\exp(163.794/(1723.261 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 2.7919 + 3.0 + 0.0 = 5.7919 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1723.261 \cdot 1.0 \cdot \ln((2 \cdot 8.0 + 56.0) / 56.0) - 0.0$$

$$= 433.08 - 0.0 = 433.0803 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c :	3.000 mm
Min. required shell thickness, external p + c :	220.145 mm

Nozzle neck minimum required thickness	: 7.520 mm
--	------------

Minimum nozzle neck thickness 10.998 mm , is greater than required thickness 7.520 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1723.3/1488.7, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 11.0 - 3.0$$

$$= 8.000 \text{ mm}$$

Nozzle Mean Radius [Rnm]:

$$= R_n + 0.5 \cdot t_n$$

$$= 28.0 + 0.5 \cdot 8.0$$

$$= 32.000 \text{ mm}$$

Parameter [λN]:

$$= 1.285 / \sqrt{(R_{nm} \cdot t_n)}$$

$$= 1.285 / \sqrt{(32.0 \cdot 8.0)}$$

$$= 0.080 \text{ 1/mm}$$

Parameter [CL]:

$$= \min(\lambda N \cdot (L_{pr1} + t + L_{pr2}), 3)$$

$$= \min(0.08 \cdot (25.0 + 537.0 + 0.0), 3)$$

$$= 3.000$$

Parameter [C1]:

$$= \sinh(CL)^2 + \sin(CL)^2$$

$$= 100.358 + 0.02$$

$$= 100.378$$

Parameter [C2]:

$$= \sinh(CL)^2 - \sin(CL)^2$$

$$= 100.358 - 0.02$$

$$= 100.338$$

Parameter [C3]:

$$= (L_{pr1} + t) / (L_{pr1} + t + \min(1/\lambda N, L_{pr2}))$$

$$= (25.0 + 537.0) / (25.0 + 537.0 + \min(1/2.04, 0.0))$$

$$= 1.000$$

Parameter [Rxn]:

$$= t_n / \ln((R_n + t_n) / R_n)$$

$$= 8.0 / \ln((28.0 + 8.0) / 28.0)$$

$$= 31.833 \text{ mm}$$

Parameter [xt]:

$$= 0.5 \cdot \lambda N (t + t_e + L_{41})$$

$$= 0.5 \cdot 0.08 (537.0 + 0.0 + 9.0)$$

$$= 21.925$$

Parameter [Ct]:

$$= \exp(-xt)$$

$$= \exp(-21.925)$$

$$= 0.000$$

Parameter [Cp]:



$$= \min(((W + 0.5 \cdot L42) / (Rn \cdot t))te, 0.6)$$

$$= \min((0.0 + 0.5 \cdot 0.0) / (28.0 \cdot 537.0) 0.0, 0.6)$$

$$= 0.000$$

Maximum Unit Moment at the Nozzle Intersection [Mo]:

$$= S \cdot trf^4 / (6 \cdot (t + Cp \cdot te)^2)$$

$$= 1488.73 \cdot 518.809^4 / (6 \cdot (537.0 + 0.0 \cdot 0.0)^2)$$

$$= 623368.438 \text{ kgf}$$

Maximum Local Primary Membrane Stress in the Nozzle [PL]:

$$= 2 \cdot Mo \cdot \lambda N^2 \cdot Rnm \cdot (CT \cdot C1 \cdot C3) / (tn \cdot C2) + P \cdot Rxn / tn$$

$$= 2 \cdot 623368 \cdot 0.08^2 \cdot 32.0 \cdot (0.0) / (8.0 \cdot 100.338) +$$

$$163.794 \cdot 31.833 / 8.0$$

$$= 651.749 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2233.10	kgf/cm ²
Local Primary Membrane Stress	PL	651.75	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lτ]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (28.0 + 8.0)$$

$$= 56.549 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 6.364, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 6.364, \text{ must be } \geq \min(0.7 \cdot 8.000, 6.000) \text{ Passed}$$

Unit Shear Force [Vs]:

$$= 0.3 \cdot S \cdot trf^4 / t^3$$

$$= 0.3 \cdot 1488.73 \cdot 518.809^4 / 537.0^3$$

$$= 2089.503 \text{ kgf/mm}$$

Thickness Value [tx1]:

$$= \min(tw1, 0.5 \cdot t)$$

$$= \min(11.0, 0.5 \cdot 537.0)$$

$$= 11.000 \text{ mm}$$

Thickness Value [tx2]:

$$= \min(\max((tw1 - 0.5 \cdot t), 0.0), 0.5 \cdot t)$$

$$= \min(\max((11.0 - 0.5 \cdot 537.0), 0.0), 0.5 \cdot 537.0)$$

$$= 0.000 \text{ mm}$$

Shear Stress [τ1]:

$$= Vs / (0.6 \cdot tx1 + 0.49 \cdot L43T)$$

$$= 2089.503 / (0.6 \cdot 11.0 + 0.49 \cdot 0.0)$$

$$= 31659.186 \text{ kgf/cm}^2$$

Shear Stress [τ2]:

$$= Vs / (0.6 \cdot tx2 + 0.49 \cdot L41T)$$

$$= 2089.503 / (0.6 \cdot 0.0 + 0.49 \cdot 6.364)$$

$$= 67007.633 \text{ kgf/cm}^2$$

Shear Stress [τ1]:

$$= 2 \cdot Mo / (t(0.6 \cdot tw1 + 0.49 \cdot L41T))$$

$$= 2 \cdot 623368 / (537.0(0.6 \cdot 11.0 + 0.49 \cdot 6.364))$$

$$= 23889.680 \text{ kgf/cm}^2$$

Shear Stress [τ2]:

$$= P / 2 \cdot Rn / (0.6 \cdot tw1 + 0.49 \cdot RL41T)$$

$$= 163.79 / 2 \cdot 28.0 / (0.6 \cdot 11.0 + 0.49 \cdot 6.364)$$

$$= 235.958 \text{ kgf/cm}^2$$



Maximum Shear Stress:
= max(τ_1 , τ_2)
= max(23889.7, 236.0)
= 23889.7 must be less than or equal to 1488.7 kgf/cm²

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 0.0508 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 565.0508 mm

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Input, Nozzle Desc: AT 2"

From: 120

Pressure for Reinforcement Calculations	P	163.794	kgf/cm ²
Temperature for Internal Pressure	Temp	300	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	300	°C

Parent Material		SA-336 F11	+IT
Parent Allowable Stress at Temperature	Sv	1488.73	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1876.18	kgf/cm ²

Outside Diameter of Bolted Blind Flange	D	3070.00	mm
Head Finished (Minimum) Thickness	t	540.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1000.0000	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	+IT
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1723.26	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.74	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm

Flange Material		SA-182 F11	
Flange Type		N/A	

Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	530.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: AT 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		8.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 56.8 \cdot (\exp(163.794/(1723.261 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 2.8318 + 3.0 + 0.0 = 5.8318 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1723.261 \cdot 1.0 \cdot \ln((2 \cdot 5.0 + 56.8) / 56.8) - 0.0$
 $= 279.456 - 0.0 = 279.4557 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c :	3.000 mm
Min. required shell thickness, external p + c :	220.145 mm

Nozzle neck minimum required thickness	: 7.520 mm
--	------------

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 7.520 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1723.3/1488.7, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 30.0 - 3.0$
 $= 27.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 8.0 - 3.0$
 $= 5.000 \text{ mm}$

Nozzle Mean Radius [Rnm]:
 $= R_n + 0.5 \cdot t_n$
 $= 28.4 + 0.5 \cdot 27.0$
 $= 41.900 \text{ mm}$

Parameter [λN]:
 $= 1.285 / \sqrt{(R_{nm} \cdot t_n)}$
 $= 1.285 / \sqrt{(41.9 \cdot 27.0)}$
 $= 0.038 \text{ 1/mm}$

Parameter [CL]:
 $= \min(\lambda N \cdot (L_{pr1} + t + L_{pr2}), 3)$
 $= \min(0.038 \cdot (1000.0 + 537.0 + 0.0), 3)$
 $= 3.000$

Parameter [C1]:
 $= \sinh(CL)^2 + \sin(CL)^2$
 $= 100.358 + 0.02$
 $= 100.378$

Parameter [C2]:
 $= \sinh(CL)^2 - \sin(CL)^2$
 $= 100.358 - 0.02$
 $= 100.338$

Parameter [C3]:
 $= (L_{pr1} + t) / (L_{pr1} + t + \min(1/\lambda N, L_{pr2}))$
 $= (1000.0 + 537.0) / (1000.0 + 537.0 + \min(1/0.97, 0.0))$
 $= 1.000$

Parameter [Rxn]:

$$= t_n / \ln((R_n + t_n) / R_n)$$

$$= 27.0 / \ln((28.4 + 27.0) / 28.4)$$

$$= 40.408 \text{ mm}$$

Parameter [xt]:

$$= 0.5 \cdot \lambda N \cdot (t + t_e + L_{41} + L_{43})$$

$$= 0.5 \cdot 0.038 \cdot (537.0 + 0.0 + 10.0 + 0.0)$$

$$= 10.449$$

Parameter [Ct]:

$$= \exp(-xt)$$

$$= \exp(-10.449)$$

$$= 0.000$$

Parameter [Cp]:

$$= \min((W + 0.5 \cdot L_{42}) / (R_n \cdot t), t_e, 0.6)$$

$$= \min((0.0 + 0.5 \cdot 0.0) / (28.4 \cdot 537.0), 0.0, 0.6)$$

$$= 0.000$$

Maximum Unit Moment at the Nozzle Intersection [Mo]:

$$= S \cdot t r f^4 / (6 \cdot (t + C_p \cdot t_e)^2)$$

$$= 1488.73 \cdot 518.809^4 / (6 \cdot (537.0 + 0.0 \cdot 0.0)^2)$$

$$= 623368.438 \text{ kgf}$$

Maximum Local Primary Membrane Stress in the Nozzle [PL]:

$$= 2 \cdot M_o \cdot \lambda N^2 \cdot R_{nm} \cdot (C_T \cdot C_1 \cdot C_3) / (t_n \cdot C_2) + P \cdot R_{xn} / t_n$$

$$= 2 \cdot 623368 \cdot 0.038^2 \cdot 41.9 \cdot (0.003) / (27.0 \cdot 100.338) +$$

$$163.794 \cdot 40.408 / 27.0$$

$$= 253.317 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2233.10	kgf/cm ²
Local Primary Membrane Stress	PL	253.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (28.4 + 27.0) / 28.4$$

$$= 1.951 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (28.4 + 27.0)$$

$$= 87.022 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 27.000, 6.000) \text{ Passed}$$

Unit Shear Force [Vs]:

$$= 0.3 \cdot S \cdot t r f^4 / t^3$$

$$= 0.3 \cdot 1488.73 \cdot 518.809^4 / 537.0^3$$

$$= 2089.503 \text{ kgf/mm}$$

Thickness Value [tx1]:

$$= \min(t_{w1}, 0.5 \cdot t)$$

$$= \min(530.0, 0.5 \cdot 537.0)$$

$$= 268.500 \text{ mm}$$

Thickness Value [tx2]:

$$= \min(\max((t_{w1} - 0.5 \cdot t), 0.0), 0.5 \cdot t)$$

$$= \min(\max((530.0 - 0.5 \cdot 537.0), 0.0), 0.5 \cdot 537.0)$$

$$= 261.500 \text{ mm}$$

Shear Stress [τ1]:

$$= V_s / (0.6 \cdot t_{x1} + 0.49 \cdot L_{43T})$$

$$= 2089.503 / (0.6 \cdot 268.5 + 0.49 \cdot 0.0)$$

$$= 1297.024 \text{ kgf/cm}^2$$

Shear Stress [τ_2]:

$$= V_s / (0.6 \cdot t_{x2} + 0.49 \cdot L_{41T})$$

$$= 2089.503 / (0.6 \cdot 261.5 + 0.49 \cdot 7.071)$$

$$= 1302.971 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= P/2(R_n + t_n) / (0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot R_{L43T})$$

$$= 163.79/2 \cdot (28.4 + 27.0) / (0.49 \cdot 7.071 + 0.6 \cdot 530.0 + 0.49 \cdot 0.0)$$

$$= 14.114 \text{ kgf/cm}^2$$

Maximum Shear Stress:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(1297.0, 1303.0, 14.1)$$

$$= 1303.0 \text{ must be less than or equal to } 1488.7 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -64 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: -20 °C
Temperature reduction per Graph 3.13	: 43 °C
The computed stress ratio is	: 0.458

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (2.292 \cdot 1.0) / (8.0 - 3.0 - 0.0)$$

$$= 0.458$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -64 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -39 °C

The Drop for this Nozzle is : 0.0508 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 1540.0508 mm

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Input, Nozzle Desc: BT 2"

From: 120

Pressure for Reinforcement Calculations	P	163.794	kgf/cm ²
Temperature for Internal Pressure	Temp	300	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	300	°C

Parent Material		SA-336 F11	+IT
Parent Allowable Stress at Temperature	Sv	1488.73	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1876.18	kgf/cm ²

Outside Diameter of Bolted Blind Flange	D	3070.00	mm
Head Finished (Minimum) Thickness	t	540.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1000.0000	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	+IT
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1723.26	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.74	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		210.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm

Flange Material		SA-182 F11	
Flange Type		N/A	

Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	530.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: BT 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		8.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 56.8 \cdot (\exp(163.794/(1723.261 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 2.8318 + 3.0 + 0.0 = 5.8318 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1723.261 \cdot 1.0 \cdot \ln((2 \cdot 5.0 + 56.8) / 56.8) - 0.0$
 $= 279.456 - 0.0 = 279.4557 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c :	3.000 mm
Min. required shell thickness, external p + c :	220.145 mm

Nozzle neck minimum required thickness	: 7.520 mm
--	------------

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 7.520 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1723.3/1488.7, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 30.0 - 3.0$
 $= 27.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 8.0 - 3.0$
 $= 5.000 \text{ mm}$

Nozzle Mean Radius [Rnm]:
 $= R_n + 0.5 \cdot t_n$
 $= 28.4 + 0.5 \cdot 27.0$
 $= 41.900 \text{ mm}$

Parameter [λN]:
 $= 1.285 / \sqrt{(R_{nm} \cdot t_n)}$
 $= 1.285 / \sqrt{(41.9 \cdot 27.0)}$
 $= 0.038 \text{ 1/mm}$

Parameter [CL]:
 $= \min(\lambda N \cdot (L_{pr1} + t + L_{pr2}), 3)$
 $= \min(0.038 \cdot (1000.0 + 537.0 + 0.0), 3)$
 $= 3.000$

Parameter [C1]:
 $= \sinh(CL)^2 + \sin(CL)^2$
 $= 100.358 + 0.02$
 $= 100.378$

Parameter [C2]:
 $= \sinh(CL)^2 - \sin(CL)^2$
 $= 100.358 - 0.02$
 $= 100.338$

Parameter [C3]:
 $= (L_{pr1} + t) / (L_{pr1} + t + \min(1/\lambda N, L_{pr2}))$
 $= (1000.0 + 537.0) / (1000.0 + 537.0 + \min(1/0.97, 0.0))$
 $= 1.000$

Parameter [Rxn]:

$$= t_n / \ln((R_n + t_n) / R_n)$$

$$= 27.0 / \ln((28.4 + 27.0) / 28.4)$$

$$= 40.408 \text{ mm}$$

Parameter [xt]:

$$= 0.5 \cdot \lambda N \cdot (t + t_e + L_{41} + L_{43})$$

$$= 0.5 \cdot 0.038 \cdot (537.0 + 0.0 + 10.0 + 0.0)$$

$$= 10.449$$

Parameter [Ct]:

$$= \exp(-xt)$$

$$= \exp(-10.449)$$

$$= 0.000$$

Parameter [Cp]:

$$= \min((W + 0.5 \cdot L_{42}) / (R_n \cdot t), t_e, 0.6)$$

$$= \min((0.0 + 0.5 \cdot 0.0) / (28.4 \cdot 537.0), 0.0, 0.6)$$

$$= 0.000$$

Maximum Unit Moment at the Nozzle Intersection [Mo]:

$$= S \cdot t r f^4 / (6 \cdot (t + C_p \cdot t_e)^2)$$

$$= 1488.73 \cdot 518.809^4 / (6 \cdot (537.0 + 0.0 \cdot 0.0)^2)$$

$$= 623368.438 \text{ kgf}$$

Maximum Local Primary Membrane Stress in the Nozzle [PL]:

$$= 2 \cdot M_o \cdot \lambda N^2 \cdot R_{nm} \cdot (C_T \cdot C_1 \cdot C_3) / (t_n \cdot C_2) + P \cdot R_{xn} / t_n$$

$$= 2 \cdot 623368 \cdot 0.038^2 \cdot 41.9 \cdot (0.003) / (27.0 \cdot 100.338) +$$

$$163.794 \cdot 40.408 / 27.0$$

$$= 253.317 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2233.10	kgf/cm ²
Local Primary Membrane Stress	PL	253.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (28.4 + 27.0) / 28.4$$

$$= 1.951 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (28.4 + 27.0)$$

$$= 87.022 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 27.000, 6.000) \text{ Passed}$$

Unit Shear Force [Vs]:

$$= 0.3 \cdot S \cdot t r f^4 / t^3$$

$$= 0.3 \cdot 1488.73 \cdot 518.809^4 / 537.0^3$$

$$= 2089.503 \text{ kgf/mm}$$

Thickness Value [tx1]:

$$= \min(t_{w1}, 0.5 \cdot t)$$

$$= \min(530.0, 0.5 \cdot 537.0)$$

$$= 268.500 \text{ mm}$$

Thickness Value [tx2]:

$$= \min(\max((t_{w1} - 0.5 \cdot t), 0.0), 0.5 \cdot t)$$

$$= \min(\max((530.0 - 0.5 \cdot 537.0), 0.0), 0.5 \cdot 537.0)$$

$$= 261.500 \text{ mm}$$

Shear Stress [τ1]:

$$\begin{aligned} &= V_s / (0.6 \cdot t_{x1} + 0.49 \cdot L_{43T}) \\ &= 2089.503 / (0.6 \cdot 268.5 + 0.49 \cdot 0.0) \\ &= 1297.024 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= V_s / (0.6 \cdot t_{x2} + 0.49 \cdot L_{41T}) \\ &= 2089.503 / (0.6 \cdot 261.5 + 0.49 \cdot 7.071) \\ &= 1302.971 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= P/2(R_n + t_n) / (0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot R_{L43T}) \\ &= 163.79/2 \cdot (28.4 + 27.0) / (0.49 \cdot 7.071 + 0.6 \cdot 530.0 + 0.49 \cdot 0.0) \\ &= 14.114 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(1297.0, 1303.0, 14.1) \\ &= 1303.0 \text{ must be less than or equal to } 1488.7 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -64 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: -20 °C
Temperature reduction per Graph 3.13	: 43 °C
The computed stress ratio is	: 0.458

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (2.292 \cdot 1.0) / (8.0 - 3.0 - 0.0) \\ &= 0.458 \end{aligned}$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -64 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -39 °C

The Drop for this Nozzle is : 0.0508 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 1540.0508 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more Nozzles failed Code Requirements!**

Please review all reports carefully!

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